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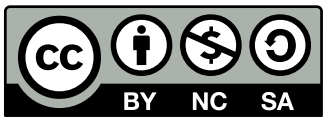
Learning Shared and Private Representations for Multi-omics Cancer Subtyping

Reza Soumi

Master's Thesis
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Abstract

Deep multi-omics models often learn shared and private latent codes for cancer subtyping and report strong unsupervised clustering metrics. This thesis asks whether those codes actually play their intended roles. On TCGA breast cancer data with multi-seed evaluation, MOCSS yields high integration NMI while private codes retain subtype structure and cross-view reconstruction gaps stay near zero. OMIDIENT appears better separated on private-subtype NMI, but probes indicate private-path collapse rather than verified privacy. Three methods are developed to enforce a clearer split: cross-view predictability constraints, joint reconstruction with VICReg independence, and spectral alignment of shared codes to a fused multi-omics patient graph. Results show that low private-subtype NMI is not a reliable disentanglement trophy, that Method 2 is preferable to the VICReg ablation of Method 1 for integration quality, and that Method 3 achieves the strongest cross-view shared-versus-private distinction among AE-private models while remaining competitive on clustering.

Keywords multi-omics, cancer subtyping, shared representation, private representation, disentanglement

Preface

This Master's thesis was carried out in the Machine Learning for Health research group at the Department of Computer Science, Aalto University. I am grateful to Professor Pekka Marttinen for supervising the work and for continuous scientific guidance on representation learning and probabilistic modelling. I thank Negar Safinianaini for co-supervision, for introducing me to the OMIDIENT line of research, and for many detailed discussions on shared–private multi-omics models and evaluation practice. Their feedback substantially shaped both the methodological choices and the critical interpretation of the experimental results.

I also thank colleagues and fellow students who provided informal feedback on figures, writing, and experimental design. Any remaining errors are my own.

Otaniemi, November 2026

Reza Soumi

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Symbols and abbreviations

Symbols

M	number of omics modalities (views)
m	modality index, $m \in \{1, \ldots, M\}$
N	number of samples (patients)
X_m	observed feature matrix of modality m
S_m	intended shared latent code of modality m
P_m	intended private / view-specific latent code of modality m
$\bar{S}$	mean shared embedding, $\bar{S} = \text{mean}_m(S_m)$
$[P; S]$	concatenation of private and shared codes
U	offline fused spectral target (Laplacian embedding)
r	spectral target dimension (here $r = 16$)
Dec, D	decoder / reconstruction network
h, f, g, k, b	Method 1 predictability networks
f_{spec}	spectral regression head mapping $\bar{S}$ to U
$\lambda_{\text{ctr}}, \lambda_{\text{vic}}, \ldots$	loss weights for contrastive, VICReg, and related terms
$\Delta\text{X}\, R^2$	cross-view probe gap $R^2(X \leftarrow S_{\text{other}}) - R^2(X \leftarrow P_{\text{other}})$
NMI	normalised mutual information
ARI	adjusted Rand index
CKA	centred kernel alignment
$\mathcal{L}$	training objective (loss)

Operators

$\| \cdot \|$	matrix or vector norm (as indicated in context)
concat$(\cdot)$	concatenation of latent codes
stopgrad$(\cdot)$	stop-gradient (detach) operator
$\mathbb{E}[\,\cdot\,]$	expectation
$R^2(Y \leftarrow Z)$	coefficient of determination of Ridge regression from Z to Y

Abbreviations

AE	autoencoder
ARI	adjusted Rand index
BRCA	breast invasive carcinoma (TCGA cohort)
CCA	canonical correlation analysis
CKA	centred kernel alignment
GRL	gradient reversal layer
InfoNCE	information noise-contrastive estimation (contrastive loss)
JIVE	Joint and Individual Variation Explained
KIRC	kidney renal clear cell carcinoma (TCGA cohort)
MLP	multilayer perceptron
MOCSS	Multi-Omics Contrastive Shared-Specific learning
MOFA	Multi-Omics Factor Analysis
NMI	normalised mutual information
OMIDIENT	multi-omics integration with Dirichlet autoencoder networks
PAM50	Prediction Analysis of Microarray 50 (breast cancer subtypes)
SNF	Similarity Network Fusion
TCGA	The Cancer Genome Atlas
UMAP	Uniform Manifold Approximation and Projection
VAE	variational autoencoder
VICReg	Variance–Invariance–Covariance Regularization

1 Introduction

Cancer is a molecularly heterogeneous disease. Patients who share a clinical diagnosis often differ in transcriptional programmes, epigenetic regulation, and post-transcriptional control, and those differences matter for prognosis and therapy selection (Hasin et al., 2017; Jiang et al., 2022). Multi-omics profiling—joint measurement of several molecular layers for the same samples—has therefore become a central resource for unsupervised discovery of tumour subtypes (Mohr et al., 2024). Among public resources, The Cancer Genome Atlas (TCGA) remains the most widely used benchmark for method development, with breast invasive carcinoma (BRCA) providing a particularly mature subtype ontology through the PAM50 labels (Parker et al., 2009).

Integrating heterogeneous omics views is not merely a matter of concatenating features. Modalities differ in dimension, noise structure, and biological content: some signals are shared across assays, while others are view-specific. Classical integrative methods such as iCluster (Shen et al., 2009), Multi-Omics Factor Analysis (MOFA) (Argelaguet et al., 2018), Joint and Individual Variation Explained (JIVE) (Lock et al., 2013), and Similarity Network Fusion (SNF) (Wang et al., 2014) already encode this intuition by separating joint from individual structure, or by fusing patient similarity graphs. Deep learning has extended the same idea to nonlinear latent spaces. Shared–private autoencoders learn, for each modality m, a shared code S_m intended to capture cross-view biology and a private code P_m intended to capture residual view-specific variation (Chen et al., 2023; Xiao et al., 2022). Contrastive alignment of projected shared codes, as in MOCSS (Chen et al., 2023), encourages cross-omics consistency, while OMIDIENT replaces the private branch with a product-Dirichlet variational model while retaining a MOCSS-style shared pathway (Safinianaini et al., 2025).

The practical success of these models is usually summarised by unsupervised clustering metrics on an integrated embedding—most often normalised mutual information (NMI) with clinical subtypes. High NMI is valuable, but it does not certify that S and P play their intended roles. A private code that quietly absorbs subtype-discriminative shared biology, or a shared code that mainly reconstructs local assay detail, can still yield competitive clustering after concatenation. Orthogonality penalties that act only through weak elementwise products do not guarantee statistical independence, and dual reconstruction paths that require both S_m and P_m to reconstruct the full input X_m actively encourage redundancy. Representation-learning research outside multi-omics has repeatedly shown that reconstruction alone does not identify the factors one cares about (Bengio et al., 2013; Goodfellow et al., 2016); the same warning applies here.

Motivating observation. Preliminary diagnostics on BRCA, including embedding visualisations of the kind later summarised in Figure 10 (Chapter 5), suggest that models with similar integration NMI can exhibit qualitatively different shared–private geometries. In MOCSS, private embeddings often retain clear subtype structure, while the shared codes look only partially aligned with the same taxonomy. In OMIDIENT, private codes appear nearly non-informative for subtypes under concatenation metrics, yet own-view reconstruction probes indicate pathological collapse rather than clean

privacy. Quantitative companions of those figures—NMI bars, metric heatmaps, and scatter plots of subtype separation against ΔX R^2—point in the same direction. These observations motivate an evaluation protocol that treats role separation as a first-class scientific claim, not as a corollary of clustering accuracy.

The thesis therefore separates two questions that are often conflated. The *clustering* question asks whether an embedding recovers PAM50 structure. The *decomposition* question asks whether cross-view signal concentrates in S while P remains residual or complementary. Answering the second requires probes that clustering tables omit.

1.1 Research questions

This thesis addresses the following questions.

1. Do high unsupervised subtype NMI scores of prior shared–private models (MOCSS, OMIDIENT) imply that S and P carry the intended complementary information on BRCA multi-omics data?

2. Which label-free probes—in particular cross-view reconstruction gaps ΔX R^2, own-view Ridge R^2, centred kernel alignment (CKA), and embedding visualisations—best diagnose failures of shared–private decomposition?

3. Can architectural and objective changes that enforce a clearer distinction between S and P (Methods 1–3 introduced in this thesis) improve role separation without sacrificing competitive integration performance?

4. Among the proposed methods, which trade-offs between predictability constraints, VICReg-style independence, joint reconstruction, and spectral alignment are empirically preferable under a multi-seed BRCA evaluation?

1.2 Objectives and scope

The objective is not to maximise subtype NMI at all costs. Instead, the thesis aims to (i) audit existing shared–private pipelines with probes that test whether foreign private codes predict another view better or worse than foreign shared codes; (ii) design Methods 1–3 that operationalise shared content as cross-view predictable structure and private content as residual, view-local variation; and (iii) compare methods under a fixed encoder backbone, label-free model selection, and multi-seed reporting. The empirical focus is TCGA BRCA with mRNA, DNA methylation, and miRNA views and PAM50 subtypes. Kidney renal clear cell carcinoma (KIRC) is included in the experimental setup as a secondary cohort; KIRC numerical results will be reported in a later revision of this thesis. Method 4 from the accompanying codebase is out of scope and is not discussed.

1.3 Contributions

The main contributions are as follows.

1. A critical multi-seed audit of MOCSS and OMIDIENT on BRCA showing that competitive integration NMI coexists with failed shared–private role assignment: MOCSS private codes steal subtype signal, while OMIDIENT private codes collapse under own-view probes despite large ΔX R^2.

2. An evaluation protocol centred on ΔX R^2, role-aware UMAP inspection, and integration NMI, with an explicit rule against ranking models by minimising concat(P) subtype NMI alone.

3. Three methods designed to enforce shared–private distinction: Method 1 (cross-view predictability with adversaries and private sufficiency), a Method 1+VICReg ablation, Method 2 (joint reconstruction with VICReg and contrastive alignment), and Method 3 (offline fused spectral targets with residual private decoding).

4. A BRCA multi-seed ($n = 5$, seeds $\{21, \ldots, 25\}$) empirical comparison identifying Method 3 as the clearest winner on ΔX R^2 while remaining competitive on concatenation NMI, and Method 2 as a simpler AE-private alternative preferred over the VICReg ablation when both role separation and integration are considered jointly.

1.4 Structure of the thesis

Chapter 2 reviews classical and deep multi-omics integration, with emphasis on shared–private models and evaluation pitfalls. Chapter 3 summarises the biological setting and the technical background on autoencoders, contrastive learning, VICReg, spectral clustering, and clustering metrics. Chapter 4 formalises the problem, describes baselines and Methods 1–3, and defines the evaluation protocol. Chapter 5 reports BRCA multi-seed results, leading with shared–private diagnostics before subtyping performance. Chapter 6 interprets the findings, limitations, and implications for future multi-omics representation learning. Chapter 7 concludes.

2 Literature review

Multi-omics integration sits at the intersection of computational biology and representation learning. This chapter reviews classical statistical approaches, deep multi-omics models, the shared–private contrastive line of work that includes MOCSS and OMIDIENT, and broader machine-learning critiques of claimed disentanglement. The chapter closes by stating the gap addressed in this thesis.

2.1 Classical integrative clustering and factorisation

Early multi-omics subtype discovery largely relied on linear latent-variable or graph-fusion models. iCluster (Shen et al., 2009) poses integrative clustering as a joint Gaussian latent-variable model: a common discrete cluster assignment generates modality-specific continuous loadings, so that subtype structure is forced to be shared while residual variation remains view-local. The method is statistically clear and remains a common baseline, but scalability and nonlinear interactions are limited.

MOFA (Argelaguet et al., 2018) and related factor models extend the idea to continuous latent factors with sparsity and likelihood choices adapted to each omics type. Factors can be shared or private depending on which views they load on, providing an interpretable decomposition without requiring a single clustering objective. JIVE (Lock et al., 2013) makes the shared–individual partition explicit at the matrix level by decomposing a multi-block data set into joint structure, individual structure, and residual noise. These methods establish the conceptual vocabulary used throughout this thesis: shared structure should explain cross-view agreement, while private structure should explain leftover within-view variance.

Graph-based approaches take a complementary route. Similarity Network Fusion (Wang et al., 2014) constructs a patient similarity network per modality and iteratively fuses them so that local neighbourhoods that agree across views are reinforced. Spectral clustering on the fused graph then yields subtypes (Ng et al., 2002; von Luxburg, 2007). SNF does not learn parametric encoders, but it demonstrates that cross-view agreement is naturally expressed as geometry on samples rather than as independent per-view reconstructions. That geometric intuition reappears in Method 3 of this thesis.

Benchmarking studies have emphasised that no single classical method dominates all cancers and metrics (Rappoport and Shamir, 2021; Cantini et al., 2021). Performance depends on preprocessing, the number of clusters, and whether the biological ground truth is itself contested. Consequently, modern work increasingly reports multiple metrics and stress-tests robustness across seeds and hyperparameter choices (Baião et al., 2025).

A further lesson from the classical literature is that "shared" has always been an operational concept. In iCluster it is the discrete assignment; in MOFA and JIVE it is the set of factors or matrix components with multi-view support; in SNF it is the fused neighbourhood relation. None of these definitions reduces to "whatever remains after training an autoencoder with an orthogonality penalty." That historical precision motivates the probe-centred evaluation used later in this thesis.

2.2 Deep multi-omics representation learning

Deep models replace linear factors with nonlinear encoders, typically autoencoders or variational autoencoders (Kingma et al., 2013; Goodfellow et al., 2016). The appeal is capacity: nonlinear maps can capture interactions among genes, CpG sites, and microRNAs that linear factor models miss. Surveys of multi-omics machine learning highlight both the promise and the risk of opaque embeddings that are hard to validate biologically (Hasin et al., 2017; Xiao et al., 2022; Jiang et al., 2022; Mohr et al., 2024). Large-scale cancer genomics programmes have accelerated the availability of matched multi-omics, but they have also increased the temptation to treat any embedding that clusters by clinical labels as scientifically resolved (Jiang et al., 2022).

Architecturally, deep integration strategies fall into early, intermediate, and late fusion. Intermediate fusion—learning per-view encoders and then combining latent codes—is the dominant pattern for unsupervised subtyping because it respects modality heterogeneity while still enabling joint clustering. Early concatenation ignores scale and noise differences; late fusion of separately clustered views can miss coordinated but weak cross-omics signal. Intermediate shared–private models attempt to keep the best of both worlds by allocating capacity to cross-view and view-local factors inside a common training loop (Xiao et al., 2022).

Contrastive learning has become a popular alignment tool in this setting: positive pairs are matched samples across views, and InfoNCE-style objectives pull their embeddings together (Chen et al., 2020). When applied to multi-omics, contrastive losses encourage cross-view consistency of the aligned subspace, but they do not by themselves define what should remain private. If the shared encoder can still reconstruct local assay detail, contrastive alignment may merely synchronise redundant copies of mixed information.

Several deep shared–private designs have been proposed. The common template encodes each view into S_m and P_m, applies a cross-view consistency loss on S, and reconstructs X_m from some combination of S_m and P_m. The details of reconstruction and independence regularisation determine whether the intended semantics hold. Weak orthogonality between S_m and P_m is often used as a cheap surrogate for disentanglement, yet elementwise product penalties do not imply subspace orthogonality or statistical independence (Zbontar et al., 2021; Bardes et al., 2022). Principal-component diagnostics of omics matrices already show that dominant variance directions need not coincide with biologically annotated structure (Yeung and Ruzzo, 2001); likewise, dominant reconstructive directions in a private code need not be "private" in the scientific sense.

2.3 MOCSS and OMIDIENT

MOCSS (Chen et al., 2023) is a representative deep shared–specific model for cancer subtyping. Each modality is mapped by an autoencoder into shared and specific latents. Two design choices are especially consequential. First, both the shared decoder and the specific decoder reconstruct the full input X_m, so each code is pressured to be individually sufficient for reconstruction. Second, shared codes

are projected by a multilayer perceptron and aligned with a contrastive (InfoNCE) loss across modalities, while a weak elementwise orthogonality term discourages co-activation of S_m and P_m. Empirically, MOCSS reports strong unsupervised subtype recovery on TCGA cohorts. The present thesis revisits that success through role-aware probes and finds that high concatenation NMI can coexist with private codes that retain—and sometimes dominate—subtype structure. Equally telling is the cross-view reconstruction pattern: predicting a modality X_A from the shared codes of the other modalities yields approximately the same Ridge R^2 as predicting X_A from those modalities' private codes, so the cross-view gap $\Delta\mathrm{X}\ R^2$ stays near zero. If S and P truly played distinct roles, foreign shared codes should systematically outperform foreign private codes at explaining another view; the near-parity of the two predictors is further evidence that the named shared and private latents are not what the architecture labels suggest.

From a modelling standpoint, dual full reconstruction is in tension with the residual interpretation of private codes. If P_m must reconstruct X_m alone, it has little incentive to leave cross-view signal exclusively to S_m. Orthogonality that penalises only per-dimension co-activation after normalisation cannot close that loophole: nonlinear mixing and subspace overlap remain possible. The contrastive term improves cross-view agreement of projected S, but agreement is compatible with both pure shared biology and synchronised leakage of private detail.

OMIDIENT (Safinianaini et al., 2025) retains a MOCSS-style shared pathway but replaces the private branch with a product-Dirichlet variational encoder–decoder. Its primary design goal is to improve the *private/specific* representation—exploiting compositional Dirichlet structure and modelling view-specific uncertainty more faithfully than a deterministic autoencoder—and the original work reports gains on clustering, classification, and missing-data reconstruction that demonstrate the value of that private-branch change. The paper's central claim is therefore about a better private model within a shared–specific template, not about establishing a new operational definition of sharedness or about proving that S and P are information-theoretically distinct. In principle, a well-behaved private Dirichlet code should absorb residual assay-specific variation while leaving cross-view subtype biology in S. When the same BRCA checkpoints are re-examined with the role probes of this thesis, multi-seed evaluation reveals near-zero concat(P) NMI together with strongly negative own-view $X \leftarrow P$ Ridge R^2. That pattern is consistent with private-path collapse under the probe suite used here, rather than with a verified shared–private theory win. The point is not that OMIDIENT lacks scientific value—on the contrary, it advances private modelling and remains a strong integration baseline—but that low private-subtype NMI alone should not be read as proof that the shared/private labels have been validated. Moreover, single-seed subtype scores that look favourable can shrink under multi-seed aggregation, underscoring the need for seed-robust reporting (Baião et al., 2025). Because OMIDIENT's shared branch is still MOCSS-like, improvements confined to the private likelihood do not automatically repair the definition of sharedness; that complementary question is the focus of the present thesis.

2.4 Shared–private learning beyond multi-omics

Outside genomics, representation learning has long distinguished factors that should be invariant across views from factors that should remain style- or domain-specific (Bengio et al., 2013; Goodfellow et al., 2016). Contrastive methods such as SimCLR (Chen et al., 2020) maximise agreement under augmentation, while non-contrastive methods such as Barlow Twins (Zbontar et al., 2021) and VICReg (Bardes et al., 2022) control redundancy through covariance and variance constraints. These ideas are transferable to multi-omics if one treats modalities as views: shared codes should agree across views, and private codes should explain residuals without duplicating the shared subspace.

Critiques of claimed disentanglement emphasise that independence penalties, reconstruction losses, and clustering metrics measure different properties (Märtens and Yau, 2024). A model can look disentangled under one probe and entangled under another. In multi-omics, an especially misleading failure mode is to treat low private-subtype NMI as proof of successful separation: private codes may be collapsed, or they may encode complementary biology that is subtype-related yet not redundant with S. Levine et al. (Levine et al., 2013) and related clinical multi-omics studies illustrate that assay-specific correlates of subtype are scientifically meaningful; discarding them by force can harm both interpretation and integration. Navigating multi-omics method choice therefore requires matching the inductive bias to the scientific question, not only to a leaderboard metric (Mohr et al., 2024).

2.5 Evaluation practice and the research gap

Technical surveys caution that preprocessing, feature selection, and metric choice can dominate apparent method rankings (Baião et al., 2025; Rappoport and Shamir, 2021; Cantini et al., 2021). Standard unsupervised scores—NMI, ARI, clustering accuracy after Hungarian matching (Kuhn, 1955), and neighbourhood accuracy—remain necessary for comparing subtype recovery, but they are silent on latent semantics. For shared–private models the evaluation gap is sharper still: standard leaderboards report clustering on concat(P, S) or on S alone, but rarely quantify whether foreign private codes predict another modality better than foreign shared codes. Without such probes, one cannot tell whether "shared" is shared.

The gap addressed by this thesis is therefore twofold. Methodologically, existing deep shared–private pipelines do not sufficiently constrain S and P to their intended roles when both codes are allowed to reconstruct X_m under weak independence regularisation. Evaluatively, high unsupervised subtype NMI has been over-interpreted as evidence of successful decomposition. The following chapters develop Methods 1–3—predictability constraints, joint reconstruction with VICReg, and spectral shared targets—and a probe-centred protocol to close that gap on BRCA, with KIRC reserved for a later revision.

3 Background

This chapter provides the biological and technical background required for the methodology. Biological subsections introduce multi-omics data, TCGA cohorts, and PAM50 subtypes. Technical subsections summarise autoencoders and VAEs, contrastive learning, VICReg, spectral clustering, and the clustering and probe metrics used later.

3.1 Multi-omics profiling in cancer

An *omics* modality is a high-dimensional molecular assay summarised as a sample-by-feature matrix. In this thesis three modalities are central: messenger RNA (mRNA) expression, DNA methylation, and microRNA (miRNA) expression. Jointly they capture transcriptional state, epigenetic regulation, and post-transcriptional control (Hasin et al., 2017). Because the assays are generated from the same tumour aliquots, matched multi-omics enables unsupervised integration aimed at patient subtypes rather than at single-assay artefacts (Jiang et al., 2022; Mohr et al., 2024).

Challenges are substantial. Feature dimensions differ by orders of magnitude before selection; noise and missingness patterns are modality-specific; and biological signal may be shared, partially shared, or private. Linear dimension reduction such as principal component analysis remains a useful preprocessing and diagnostic tool (Yeung and Ruzzo, 2001), but nonlinear encoders are often needed once interactions across thousands of features are considered (Goodfellow et al., 2016).

3.2 TCGA BRCA and KIRC

The Cancer Genome Atlas provides harmonised multi-omics for many tumour types. Breast invasive carcinoma (BRCA) is the primary cohort in this work. After the preprocessing used in the accompanying codebase, BRCA comprises $N = 875$ samples with 1000 mRNA features, 1000 methylation features, and 503 miRNA features, labelled into five PAM50 subtypes (Parker et al., 2009). Kidney renal clear cell carcinoma (KIRC) is a secondary cohort with $N = 289$ samples, much higher raw feature dimensionality (58316 mRNA, 22928 methylation, 1879 miRNA features in the prepared tables), and two clinical classes in the evaluation setting used here. KIRC results are deferred to a later revision; the cohort is mentioned so that the experimental design remains explicit about intended generalisation beyond BRCA.

3.3 PAM50 breast cancer subtypes

PAM50 is a nearest-centroid classifier based on a fifty-gene expression signature that assigns breast tumours to intrinsic subtypes such as Luminal A, Luminal B, HER2-enriched, Basal-like, and Normal-like (Parker et al., 2009). Although derived from transcriptomics, PAM50 labels correlate with multi-omics structure and are the de facto unsupervised benchmark target for BRCA integration methods (Chen et al., 2023; Rappoport and Shamir, 2021). In this thesis PAM50 labels are used only for

evaluation, never for model selection: checkpoints are ranked by label-free validation loss.

3.4 Autoencoders and variational autoencoders

An autoencoder (AE) learns an encoder f_{enc} and decoder f_{dec} by minimising a reconstruction loss between X and $\hat{X} = f_{\text{dec}}(f_{\text{enc}}(X))$ (Goodfellow et al., 2016). Bottleneck latents act as compressed representations for downstream clustering. A variational autoencoder (VAE) places a probabilistic encoder $q(z \mid x)$ and a prior $p(z)$ in an evidence lower-bound objective (Kingma et al., 2013). OMIDIENT's private branch is a Dirichlet-oriented VAE variant rather than a standard Gaussian VAE (Safinianaini et al., 2025). Throughout Methods 1–3, unless otherwise stated, private and shared encoders follow a deterministic AE parameterisation aligned with MOCSS for fair comparison.

3.5 Contrastive learning and InfoNCE

Contrastive learning trains embeddings so that positive pairs are close and negatives are far in a projected space (Chen et al., 2020). The InfoNCE loss is a standard instantiation: for a batch of projected embeddings, each anchor is classified against negatives via a temperature-scaled softmax. In multi-omics shared–private models, positives are matched samples' shared codes across modalities. Contrastive pressure encourages cross-view alignment of S, but if reconstruction also forces S_m to encode private detail, contrastive learning alone cannot purify the shared subspace.

3.6 VICReg and independence regularisation

Variance–Invariance–Covariance Regularization (VICReg) (Bardes et al., 2022) stabilises non-contrastive learning through three terms: a variance hinge that prevents dimensional collapse, a covariance penalty that reduces redundancy within an embedding, and—when applied to two embeddings—a cross-covariance penalty that encourages linear independence. Barlow Twins pursues a related redundancy-reduction goal (Zbontar et al., 2021). In this thesis a VICReg-style penalty is applied between P_m and S_m inside each modality for Method 1+VICReg and Method 2, replacing MOCSS's weak elementwise orthogonality. Linear independence is not full statistical disentanglement (Märtens and Yau, 2024), but it is a stricter and more interpretable constraint than elementwise co-activation penalties.

3.7 Spectral clustering and fused graphs

Spectral clustering embeds samples using eigenvectors of a graph Laplacian and then applies k-means in that embedding (Ng et al., 2002; von Luxburg, 2007; Lloyd, 1982). SNF-style fusion builds a single patient graph that aggregates neighbourhood agreement across modalities (Wang et al., 2014). Method 3 precomputes a fused k-nearest-neighbour graph offline, extracts a Laplacian embedding $U \in \mathbb{R}^{N \times r}$ with

$r = 16$, and trains shared codes to regress toward U. The NMI of k-means on U itself is an unsupervised ceiling for how much subtype signal the spectral target contains; on BRCA that ceiling is approximately $\mathrm{NMI}(U) \approx 0.387$ in the configuration used here.

3.8 Clustering metrics and assignment

Unsupervised subtype recovery is evaluated with normalised mutual information (NMI), adjusted Rand index (ARI), clustering accuracy after optimal label permutation via the Hungarian algorithm (Kuhn, 1955), and k-nearest-neighbour classification accuracy in the embedding space. NMI and ARI compare partitions without requiring label alignment; accuracy requires a matching between cluster indices and true labels. All clustering in this thesis uses k-means with $n_{\text{init}} = 30$ (Lloyd, 1982).

3.9 Probe metrics for shared–private roles

Because clustering metrics do not test role assignment, additional probes are required. Cross-view Ridge regression asks how well modality A's features are predicted from other modalities' shared codes versus private codes. The signed gap

$$\Delta\mathrm{X}\, R^2 \;=\; R^2(X_A \leftarrow (S_B, S_C)) - R^2(X_A \leftarrow (P_B, P_C)) \tag{1}$$

is positive when foreign shared codes explain X_A better than foreign private codes—the intended asymmetry. Own-view Ridge R^2 for $X_m \leftarrow S_m$, $X_m \leftarrow P_m$, and $X_m \leftarrow [P_m; S_m]$ diagnoses collapse and redundancy within a modality. Centred kernel alignment (CKA) summarises linear similarity among shared codes across views. Embedding visualisations (UMAP) provide qualitative evidence of whether subtype geometry lives primarily in S, in P, or in both.

Interpretation rule. Low concat(P) subtype NMI is not by itself evidence of good disentanglement: private codes may have collapsed, or they may carry complementary subtype-related biology. Models are therefore judged jointly by $\Delta\mathrm{X}\, R^2$, UMAP role separation, and integration NMI on concat(P, S) or related fused embeddings.

4 Methodology

This chapter formalises shared–private multi-omics learning, describes the MOCSS and OMIDIENT baselines, introduces Methods 1–3, and defines the evaluation protocol with emphasis on ΔX R^2.

4.1 Problem formalisation

Let $M = 3$ matched omics modalities be observed for N samples. For modality $m \in \{1, 2, 3\}$, write $X_m \in \mathbb{R}^{N \times d_m}$. A shared–private encoder produces

$$(S_m, P_m) = E_m(X_m), \qquad S_m, P_m \in \mathbb{R}^{N \times d}, \tag{2}$$

with embedding dimension $d = 32$ in all experiments. Intended semantics are:

- S_m should capture information about sample i that is predictable across modalities (cross-view biology, including subtype-related programmes);
- P_m should capture residual view-specific variation useful for reconstructing X_m but not for explaining other views.

Downstream unsupervised subtyping is performed by k-means on $\text{mean}_m(S_m)$, on $\text{concat}_m(P_m)$, and on $\text{concat}(P_1, P_2, P_3, \text{mean}_m(S_m))$, among other probes. The scientific claim of interest is not only that some embedding clusters by subtype, but that S and P realise the decomposition above.

Shared implementation conventions. Unless a method states otherwise, encoders are four-layer multilayer perceptrons with tanh activations and widths [512, 256, 128, 32]. Decoders and predictors use four-layer MLPs with tanh hidden units and a final sigmoid. Shared codes are passed through a two-layer tanh projection before contrastive learning. Weights are Kaiming-initialised with zero biases. Reconstruction losses follow the MOCSS-style centred, row-normalised matrix-norm metric rather than a literal mean squared error, ensuring fair comparison with the baselines. The contrastive term is the sum of pairwise InfoNCE losses among the three projected shared embeddings in a batch (Chen et al., 2020, 2023).

4.2 Baseline: MOCSS

MOCSS (Chen et al., 2023) trains, for each modality, dual reconstruction paths

$$\hat{X}_m^{(S)} = \text{Dec}_S(S_m), \qquad \hat{X}_m^{(P)} = \text{Dec}_P(P_m), \tag{3}$$

together with a weak elementwise orthogonality penalty between ℓ_2-normalised S_m and P_m, and InfoNCE on $\text{MLP}(S_m)$ across modalities. Dual full reconstruction is the central inductive bias under scrutiny: each code is encouraged to be individually sufficient for X_m, which conflicts with a residual interpretation of P_m and invites private codes to store shared subtype signal.

4.3 Baseline: OMIDIENT

OMIDIENT (Safinianaini et al., 2025) keeps a MOCSS-style shared autoencoder and contrastive head, but replaces the private pathway with a product-Dirichlet variational encoder–decoder and a corresponding KL regulariser. The hope is that compositional private latents better absorb view-specific residual structure. The shared branch, however, remains essentially MOCSS. This thesis therefore treats OMIDIENT as a private-branch intervention whose shared–private claims must still be verified with the same probes used for AE-private models. A Method 2 hybrid that grafts Dirichlet private codes onto Method 2's joint reconstruction objective is included as a diagnostic companion.

4.4 Method 1: predictability surrogates

Method 1 replaces dual solo reconstruction as the definition of shared versus private with a set of predictability constraints. For target modality A and the other two modalities $\{B, C\}$, five networks are trained:

$$h_A(P_A, S_A) \to X_A \qquad \text{(within-view joint reconstruction)}, \tag{4}$$
$$f_A(S_B, S_C) \to X_A \qquad \text{(cross-shared sufficiency)}, \tag{5}$$
$$g_A(P_B, P_C) \to X_A \qquad \text{(cross-private adversary)}, \tag{6}$$
$$k_A(S_A) \to X_A \qquad \text{(own-shared insufficiency)}, \tag{7}$$
$$b_A(P_A) \to X_A \qquad \text{(own-private sufficiency)}. \tag{8}$$

Shared codes are additionally aligned by InfoNCE on MLP(S). Figure 1 summarises the architecture.

Let $\mathcal{L}_{\text{std}}$, $\mathcal{L}_{\text{shared}}$, $\mathcal{L}_{\text{adv}}$, $\mathcal{L}_{\text{own}}$, and $\mathcal{L}_{\text{priv}}$ denote the sums over modalities of the corresponding reconstruction metrics. Training uses a two-step procedure per batch. Step 1 updates only the adversaries g and k on detached codes:

$$\mathcal{L}_{\text{step1}} = \sum_A \left(\mathcal{L}^A_{\text{adv}} + \mathcal{L}^A_{\text{own}}\right). \tag{9}$$

Step 2 updates encoders and constructive predictors with adversaries frozen:

$$\mathcal{L}_{\text{step2}} = \sum_A \left[\mathcal{L}^A_{\text{std}} + \alpha\, \mathcal{L}^A_{\text{shared}} + \lambda_{\text{priv}}\, \mathcal{L}^A_{\text{priv}} - \beta\, \mathcal{L}^A_{\text{adv}} - \gamma\, \mathcal{L}^A_{\text{own}}\right] + \lambda_{\text{ctr}}\, \mathcal{L}_{\text{ctr}}. \tag{10}$$

The minus signs encourage encoders to make foreign private codes uninformative about X_A and to make own S_A insufficient for full reconstruction, while b_A prevents the trivial collapse of P_A under privacy pressure. Checkpointing uses a constructive validation score that drops the signed adversarial terms, because raw $\mathcal{L}_{\text{step2}}$ can become negative and unstable as an early-stopping signal. Across configurations, models are ranked by the label-free core validation loss $\mathcal{L}_{\text{std}} + \mathcal{L}_{\text{shared}} + \mathcal{L}_{\text{priv}} + \mathcal{L}_{\text{ctr}}$.

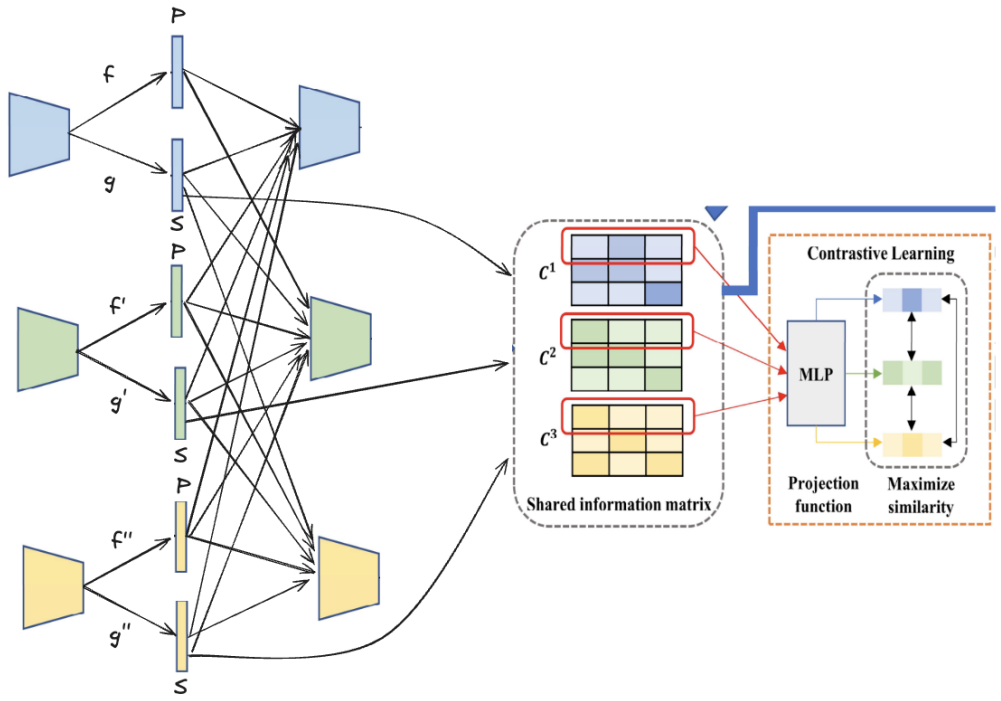


Figure 1: Method 1 architecture. Per-modality encoders emit (P_m, S_m). Predictors h, f, g, k, b implement joint reconstruction, cross-shared sufficiency, private non-leakage, own-shared insufficiency, and private sufficiency. A contrastive head aligns projected shared codes.

4.5 Method 1 with VICReg ablation

The incentive for this ablation is practical and diagnostic. Full Method 1 couples five predictors per modality with a two-step minimax procedure, and the own-shared adversary k (weight γ) is particularly delicate: too strong a γ can strip useful shared content from S_A, while too weak a γ leaves private bleed in S. The Method 1+VICReg ablation therefore asks whether the *cross-view* predictability story (h, f, g, and private sufficiency b) already suffices when supplemented by a modern linear independence regulariser, without the brittle own-shared insufficiency term. Concretely, it removes k and adds a VICReg-style independence penalty between P_m and S_m (Bardes et al., 2022). Predictors h, f, g, b remain. The VICReg term combines variance hinges on both codes, within-code covariance whitening, and a cross-covariance penalty that targets linear $P \perp S$ independence. Figure 2 illustrates the resulting architecture. Two-step training reduces to an adversary step on g only. Ranking again uses $\mathcal{L}_{\text{std}} + \mathcal{L}_{\text{shared}} + \mathcal{L}_{\text{priv}} + \mathcal{L}_{\text{ctr}}$, excluding $\mathcal{L}_{\text{vic}}$ from the core rank so that independence regularisation is treated as a training regulariser rather than as the selection criterion. Empirically (Chapter 5), the ablation is more stable than full Method 1 and produces a cleaner subtype NMI ordering of S over P, but it still fails the cross-view $\Delta X\, R^2$ test—which motivates the simpler Method 2 and the spectral Method 3 that follow.

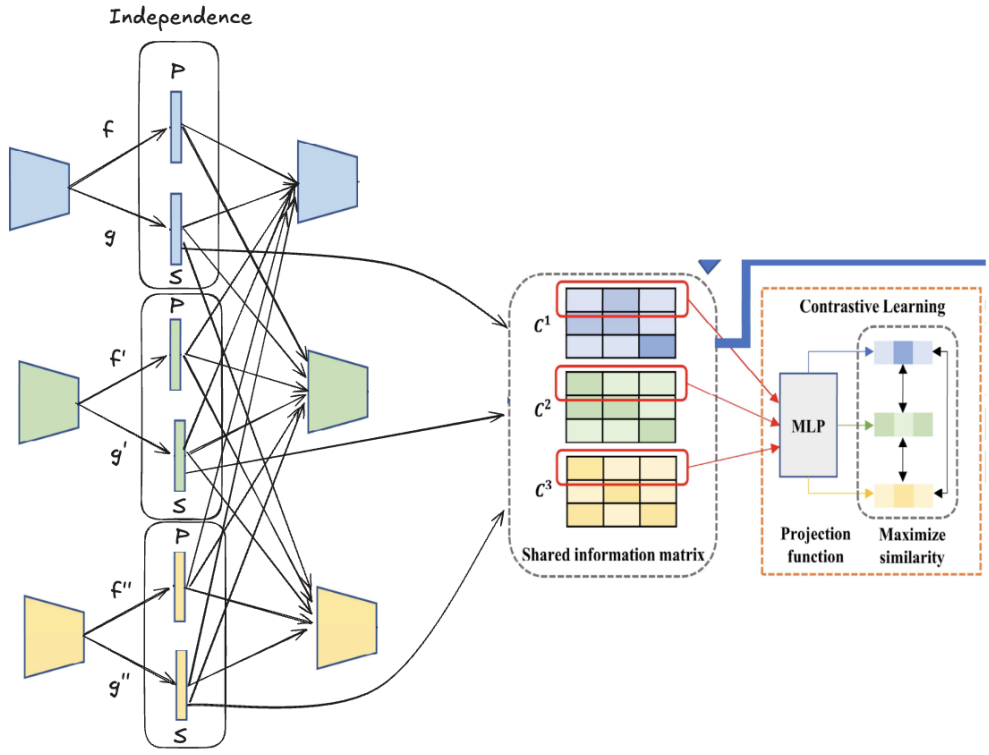


Figure 2: Method 1+VICReg ablation. The own-shared adversary k is removed; a VICReg $P \perp S$ term is added while retaining cross-view predictors h, f, g and private sufficiency b.

4.6 Method 2: joint reconstruction and VICReg

Method 2 abandons cross-reconstruction adversaries in favour of a simpler inductive bias. Each modality is reconstructed only from the concatenation of its own codes,

$$\hat{X}_m = D_m\big([P_m; S_m]\big), \tag{11}$$

while VICReg encourages $P_m \perp S_m$ and InfoNCE aligns projected shared codes (Figure 3). The training objective is

$$\mathcal{L} = \lambda_{\text{rec}}\,\mathcal{L}_{\text{recon}} + \lambda_{\text{ctr}}\,\mathcal{L}_{\text{ctr}} + \lambda_{\text{vic}}\,\mathcal{L}_{\text{vic}}, \tag{12}$$

with all terms non-negative, so raw validation loss is safe for early stopping. Configurations are ranked by $\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{ctr}}$. Relative to MOCSS, Method 2 removes the requirement that S_m or P_m alone reconstruct X_m, which is the most direct architectural fix of the dual-reconstruction failure mode.

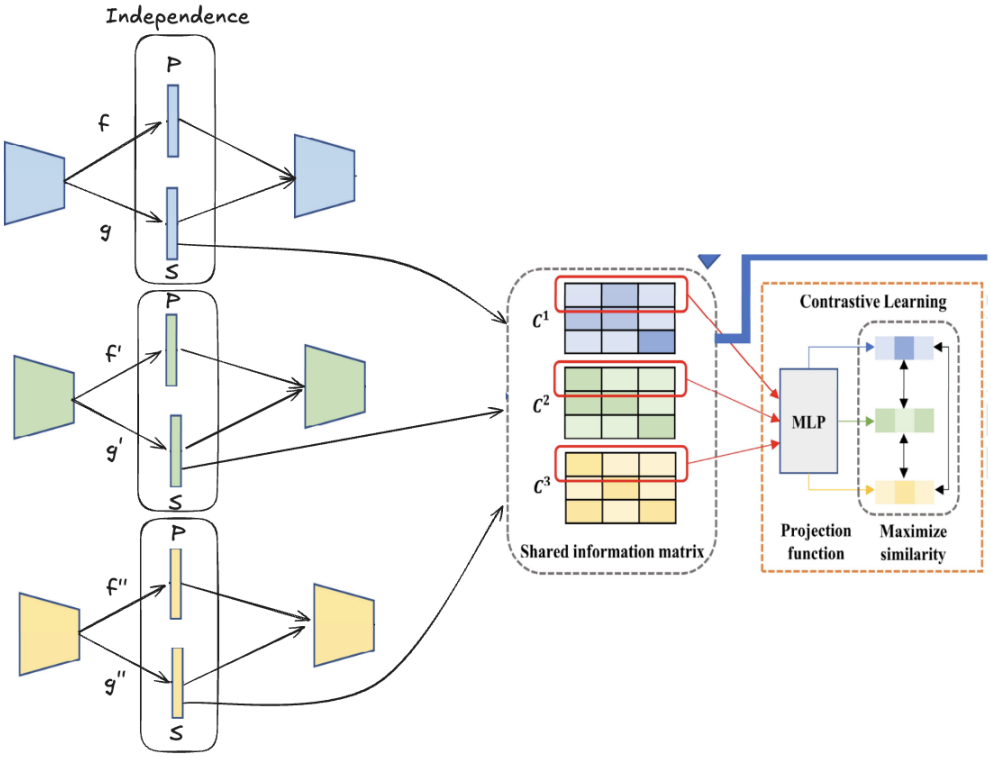


Figure 3: Method 2 architecture. A single joint decoder reconstructs X_m from $[P_m; S_m]$; VICReg replaces elementwise orthogonality; contrastive learning aligns shared codes.

A companion variant, Method 2+OMIDIENT-private, replaces only the private encoder with OMIDIENT's product-Dirichlet branch while retaining Method 2's joint decoder, VICReg, and contrastive shared pathway. It is used to test whether Dirichlet private codes remain well-behaved once dual solo reconstruction is removed.

4.7 Method 3: spectral shared targets with residual private decoding

Method 3 supplies an explicit clustering-shaped target for shared codes. Offline, modality-wise k-nearest-neighbour graphs are fused and a normalised Laplacian embedding $U \in \mathbb{R}^{N \times r}$ with $r = 16$ is computed (Wang et al., 2014; Ng et al., 2002; von Luxburg, 2007). A spectral head f_{spec} maps $\bar{S} = \text{mean}_m(S_m)$ toward U:

$$\mathcal{L}_{\text{spec}} = \frac{1}{N} \sum_{i=1}^{N} \left\| f_{\text{spec}}(\bar{S}_i) - U_i \right\|_2^2. \tag{13}$$

Private codes reconstruct residuals through

$$\hat{X}_m = \text{Dec}_m\big(P_m, \text{stopgrad}(S_m)\big), \tag{14}$$

so gradients from residual decoding do not rewrite the shared code as a second full autoencoder of X_m. Additional terms include a view-identity head on P (encouraging assay-specific content to remain in the private branch), optional VICReg, a gradient-reversal (GRL) leakage adversary that discourages private codes from predicting soft clusters of $\bar{S}$, and a contrastive term whose weight λ_{ctr} is often selected as 0 under loss-based ranking. Figure 4 shows the architecture. Core ranking uses $\mathcal{L}_{\text{spec}} + \mathcal{L}_{\text{res}} + \mathcal{L}_{\text{ctr}}$. The unsupervised NMI of k-means on U itself, $\text{NMI}(U) \approx 0.387$ on BRCA, is reported as a spectral ceiling. Full notation and formulae for the offline spectral target, spectral regression, view head, and GRL non-leakage block are collected in Appendix A.

4.8 Training protocol

All BRCA runs use batch size 32, up to 100 epochs, and early stopping with patience 20 on the method-specific validation score (min_delta = 0.005). Learning rates and weight decays follow the MOCSS defaults used in the repository unless a sweep token indicates otherwise. Model selection is label-free: PAM50 labels never enter the ranking loss. After selection, clustering and probes are evaluated on the chosen configuration and aggregated over training seeds $\{21, 22, 23, 24, 25\}$.

4.9 Evaluation protocol and interpretation of ΔX R^2

The evaluation suite comprises:

1. subtype NMI, ARI, accuracy, and kNN accuracy on $\text{mean}(S)$, $\text{concat}(P)$, and $\text{concat}(P, S)$;
2. cross-view ΔX R^2 and related Ridge probes;
3. own-view Ridge R^2 for $X \leftarrow S$, $X \leftarrow P$, and $X \leftarrow [P; S]$;
4. pairwise CKA among shared codes;
5. UMAP visualisations of $\text{mean}(S)$ and $\text{concat}(P)$.

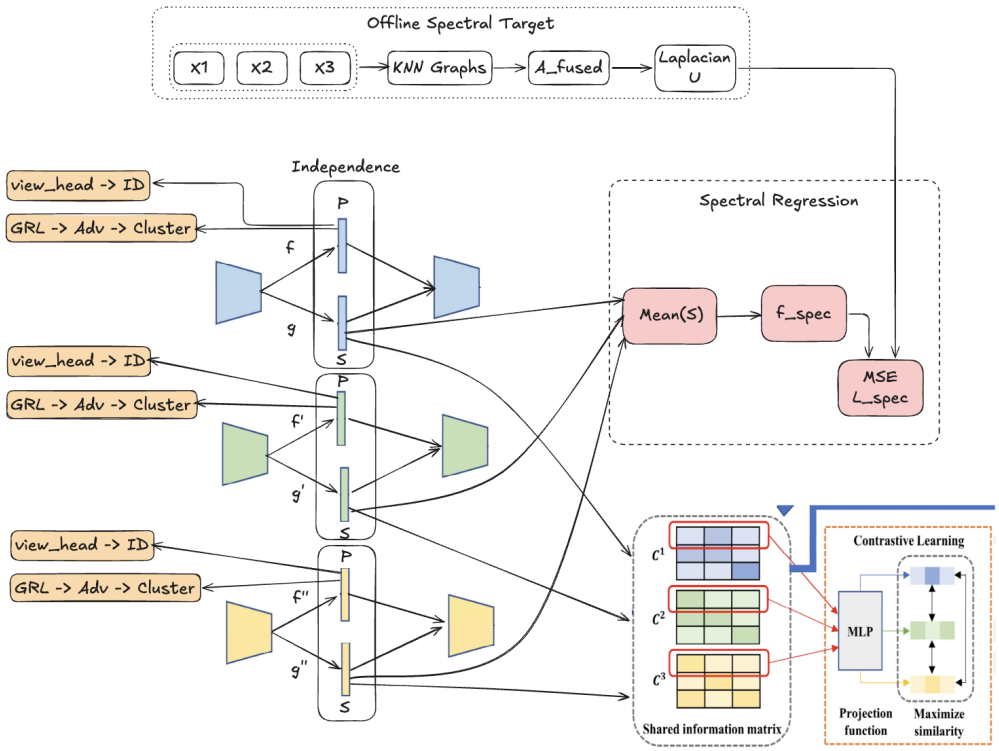


Figure 4: Method 3 architecture. An offline fused spectral embedding U supervises mean shared codes via f_{spec}; private decoders reconstruct residuals with stop-gradient on S; optional view-head and GRL leakage controls regularise role separation.

How to read $\Delta X\, R^2$. A reliably positive gap means foreign shared codes explain another modality better than foreign private codes—evidence that cross-view signal concentrates in S. Near-zero gaps indicate indistinct roles. Large positive gaps must still be checked against own-view $X \leftarrow P$: if that quantity is strongly negative, private codes have collapsed and the gap is not evidence of healthy privacy. Likewise, moderately high concat(P) NMI is acceptable when UMAP layouts differ from those of S and $\Delta X\, R^2$ remains positive; private codes may then carry complementary subtype correlates rather than stolen shared geometry (Levine et al., 2013).

Optional integration gap. An auxiliary quantity $\text{gap}_{S-\text{PS}} = \text{mean}(S)\,\text{NMI} - \text{concat}(P, S)\,\text{NMI}$ asks whether adding private codes hurts or helps the fused clustering embedding. A large positive gap means private content dilutes an already subtype-rich shared code; a negative gap means private codes contribute additional subtype structure to the concatenation. This thesis treats $\text{gap}_{S-\text{PS}}$ as secondary: it is useful context in compact tables, but it does not by itself certify role separation. The primary decomposition probes remain $\Delta X\, R^2$, own-view $X \leftarrow P$, and the comparative UMAP geometries of S and P.

4.10 Strengths and limitations

Table 1 summarises qualitative trade-offs. Method 1 offers the most explicit information-theoretic story but is heavy: fifteen predictors, two optimisers, and sensitive $(\alpha, \beta, \gamma, \lambda_{\text{priv}})$ knobs. Method 1+VICReg reduces adversary complexity and adds linear independence, yet removing k weakens the insufficiency constraint on own S. Method 2 is stable and simple, fixing MOCSS's dual-reconstruction flaw directly, but relies on VICReg and contrastive weights to prevent asymmetric collapse. Method 3 provides the strongest clustering-shaped shared target and the clearest empirical $\Delta X\, R^2$ gains, at the cost of dependence on the quality of the offline fused graph and a spectral ceiling below the best concatenation NMI of reconstruction-based models.

Table 1: Qualitative comparison of methods studied in this thesis (AE-private line unless noted).

	Reconstruction	Independence / privacy	Main risk
MOCSS	dual solo Dec_S, Dec_P	weak elementwise ortho	P steals shared subtype
OMIDIENT	Dirichlet private + MOCSS shared	ortho optional; KL on P	private-path collapse
Method 1	h, f, g, k, b predictability	adversaries + b	minimax instability; weak concat NMI
M1+VICReg	h, f, g, b + no k	VICReg $P \perp S$	low concat$(P) \neq$ disentangling
Method 2	joint $D([P; S])$	VICReg + contrastive	needs careful λ balance
Method 3	residual $\text{Dec}(P, \text{stopgrad}(S))$	spectral U + GRL leak	limited by NMI(U) ceiling

5 Results

This chapter reports empirical findings on TCGA BRCA. Shared–private diagnostics are presented before subtyping leaderboards, reflecting the thesis claim that integration NMI alone is an incomplete success criterion. All headline numbers are mean ± sample standard deviation over five training seeds $\{21, 22, 23, 24, 25\}$ unless noted. KIRC is described in the setup but numerical KIRC results are deferred to a later revision.

5.1 Experimental setup

Datasets. Table 2 summarises the prepared cohorts. BRCA is the evaluation focus: three modalities after feature selection, $N = 875$ samples, and five PAM50 subtypes (Parker et al., 2009). KIRC is included in the repository pipeline with two evaluation classes; its results will be reported in a later revision of this thesis.

Table 2: Dataset statistics for the prepared multi-omics tables used in this thesis.

Cohort	mRNA	Methylation	miRNA	Samples	Subtypes / classes
BRCA	1000	1000	503	875	5 (PAM50)
KIRC	58316	22928	1879	289	2

Training and selection. Encoders are four-layer tanh MLPs with widths $[512, 256, 128, 32]$. BRCA models use batch size 32, at most 100 epochs, and early stopping as described in Chapter 4. Hyperparameter grids are method-specific; the reported "loss-best" row for each method is the configuration with best label-free core validation loss on the development protocol, then re-evaluated across seeds. Clustering uses k-means with $n_{\text{init}} = 30$ (Lloyd, 1982). Hungarian matching is used for clustering accuracy (Kuhn, 1955).

Compared systems. AE-private comparisons include MOCSS (Chen et al., 2023), Method 1+b_m, Method 1+VICReg+b_m, Method 2, and Method 3. The OMIDIENT family includes OMIDIENT (Safinianaini et al., 2025) and Method 2 with OMIDIENT private codes. Method 4 from the codebase is not evaluated here.

5.2 Shared versus private analysis

The central empirical claim is that competitive subtype NMI does not imply successful shared–private decomposition. Figures 5–10 and Tables 3–4 support that claim.

MOCSS: private codes steal subtype. For MOCSS, concat(P) NMI (0.466 ± 0.017) is essentially tied with concat(P, S) (0.461 ± 0.008) and exceeds mean(S) (0.418 ± 0.014). The cross-view gap is only $\Delta\text{X}\, R^2 = 0.045 \pm 0.023$, indicating weak preference for foreign shared codes over foreign private codes. Own-view Ridge scores reinforce the redundancy story: $X \leftarrow P$ (0.481 ± 0.006) exceeds $X \leftarrow S$

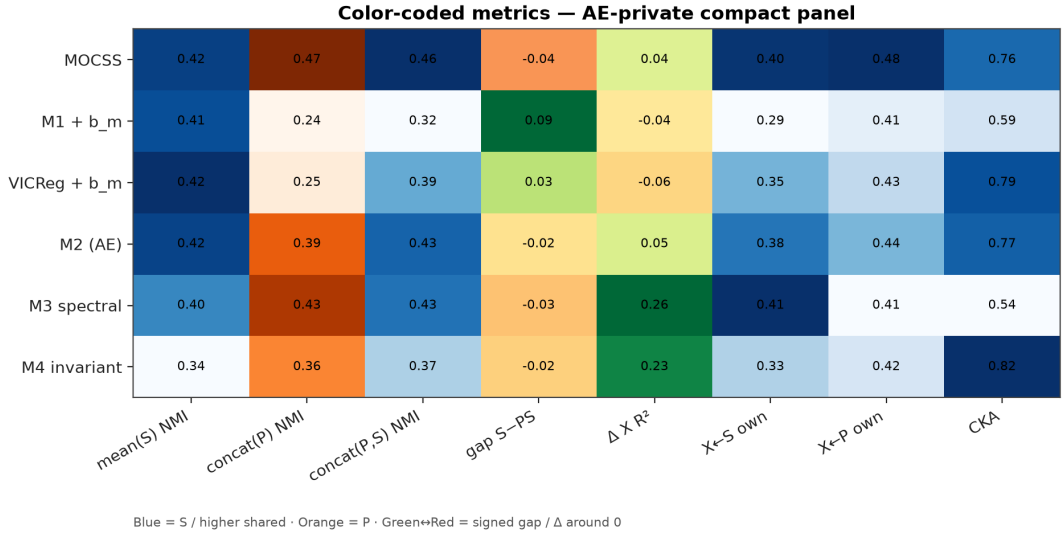


Figure 5: Compact metric heatmap for AE-private methods on BRCA (multi-seed means). Rows are methods; columns include mean(S) NMI, concat(P) NMI, concat(P, S) NMI, ΔX R^2, own-view probes, and CKA. Colour encodes relative magnitude within each column.

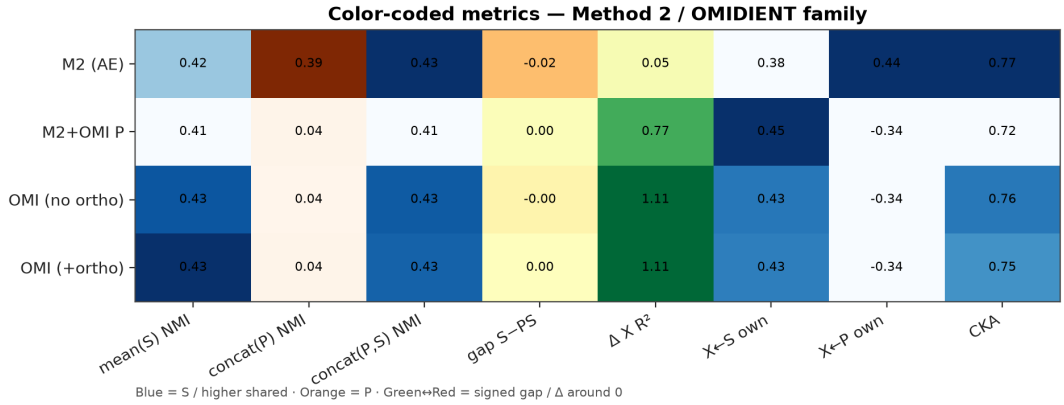


Figure 6: Compact metric heatmap for the OMIDIENT family on BRCA. Near-zero concat(P) NMI co-occurs with large ΔX R^2 and pathological own-view $X \leftarrow P$ scores.

Table 3: Shared–private diagnostic panel on BRCA (AE private). Mean ± std over seeds $\{21, \ldots, 25\}$.

Method	mean(S) NMI	concat(P) NMI	concat(P, S) NMI	ΔX R^2
MOCSS	0.418 ± 0.014	0.466 ± 0.017	0.461 ± 0.008	0.045 ± 0.023
M1+b_m	0.414 ± 0.030	0.237 ± 0.099	0.323 ± 0.063	-0.038 ± 0.095
M1+VICReg+b_m	0.423 ± 0.021	0.254 ± 0.051	0.394 ± 0.035	-0.061 ± 0.071
Method 2	0.417 ± 0.028	0.389 ± 0.034	0.434 ± 0.033	0.051 ± 0.044
Method 3	0.414 ± 0.029	0.443 ± 0.014	0.456 ± 0.025	0.248 ± 0.023

Table 4: Shared–private diagnostic panel on BRCA (OMIDIENT family). Mean ± std over seeds $\{21, \ldots, 25\}$.

Method	mean(S) NMI	concat(P) NMI	concat(P, S) NMI	ΔX R^2
OMIDIENT	0.426 ± 0.046	0.039 ± 0.009	0.430 ± 0.054	1.114 ± 0.013
M2+OMID. private	0.410 ± 0.018	0.043 ± 0.002	0.409 ± 0.014	0.770 ± 0.055

(0.404 ± 0.013), so private codes are highly reconstructive of the full input. Figure 7 visualises the NMI pattern; Figure 10 shows subtype geometry clearly present in MOCSS private embeddings. High unsupervised integration NMI therefore does *not* imply that S and P play distinct roles.

OMIDIENT: collapse rather than verified privacy. OMIDIENT yields mean(S) NMI 0.426 ± 0.046 and concat(P) NMI 0.039 ± 0.009, with a very large $\Delta X\, R^2$ = 1.114 ± 0.013. Read in isolation, these numbers look like extreme shared–private separation. Own-view probes contradict that reading: $X \leftarrow P \approx -0.342$ is strongly negative, consistent with private-path collapse of the Dirichlet branch rather than informative privacy. Single-seed shared NMI near 0.50 shrinks under multi-seed aggregation, cautioning against leaderboard optimism from one seed (Baião et al., 2025). Method 2+OMIDIENT-private inherits the same private collapse signature ($X \leftarrow P \approx -0.341$) despite Method 2's joint decoder, showing that Dirichlet private geometry—not only MOCSS dual reconstruction—can drive the failure.

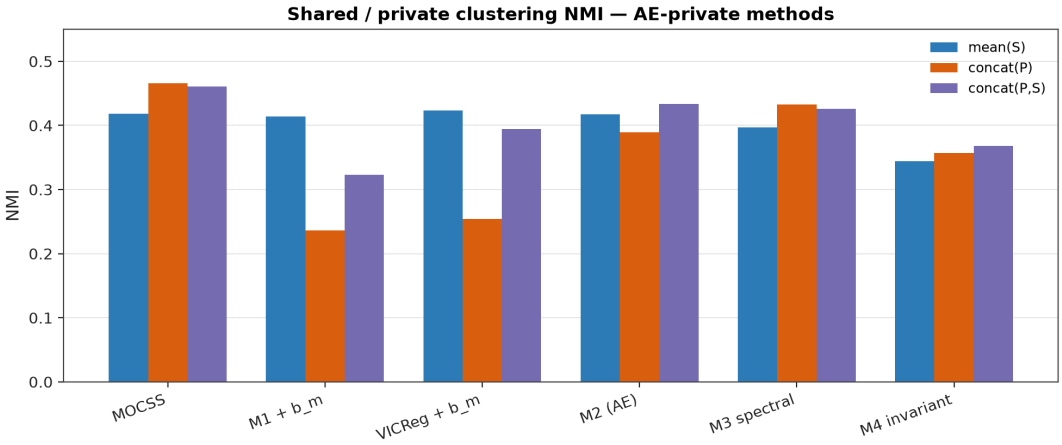


Figure 7: BRCA multi-seed NMI bars for AE-private methods: mean(S), concat(P), and concat(P, S). MOCSS private NMI rivals concatenation; Method 1 variants suppress private NMI but also weaken concatenation; Methods 2–3 preserve competitive concatenation.

Method 1 and Method 1+VICReg. Method 1+b_m reduces concat(P) NMI to 0.237 ± 0.099, but concatenation NMI falls to a weak 0.323 ± 0.063 and $\Delta X\, R^2$ is noisy and on average slightly negative (−0.038 ± 0.095). These results are not promising as a shared–private solution for two related reasons. First, the drop in concat(P) NMI was bought at the price of integration: the fused embedding used for subtyping becomes substantially worse than MOCSS (0.323 vs. 0.461), so the model no longer delivers competitive unsupervised subtyping even if private subtype content has been partially suppressed. Second, the intended cross-view asymmetry still fails: $\Delta X\, R^2$ does not become stably positive, and seed variance is large (±0.095), which is consistent with the brittleness of Method 1's two-step adversaries. In short, Method 1

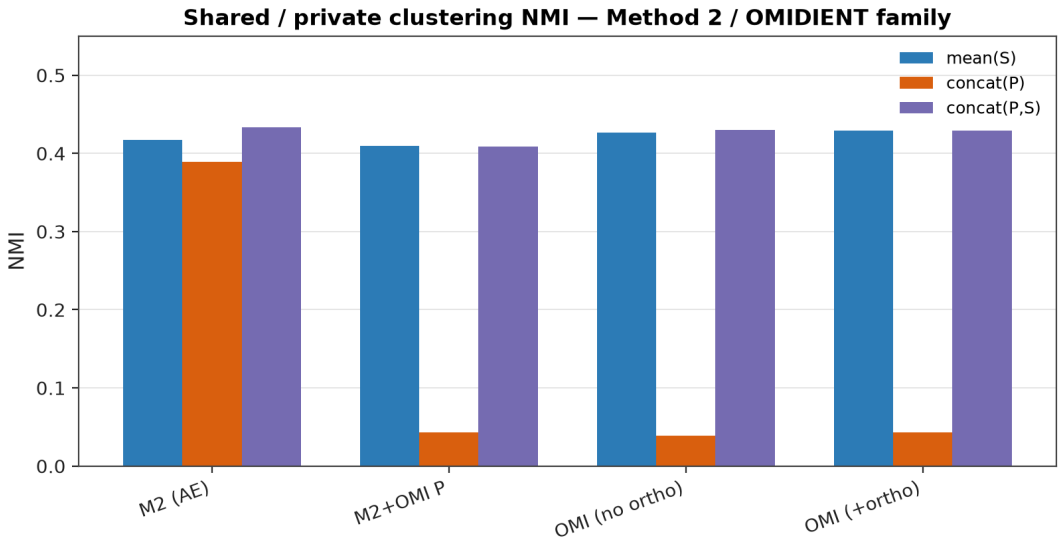


Figure 8: BRCA multi-seed NMI bars for the OMIDIENT family. Near-zero concat(P) NMI should not be read as successful disentanglement without own-view probes.

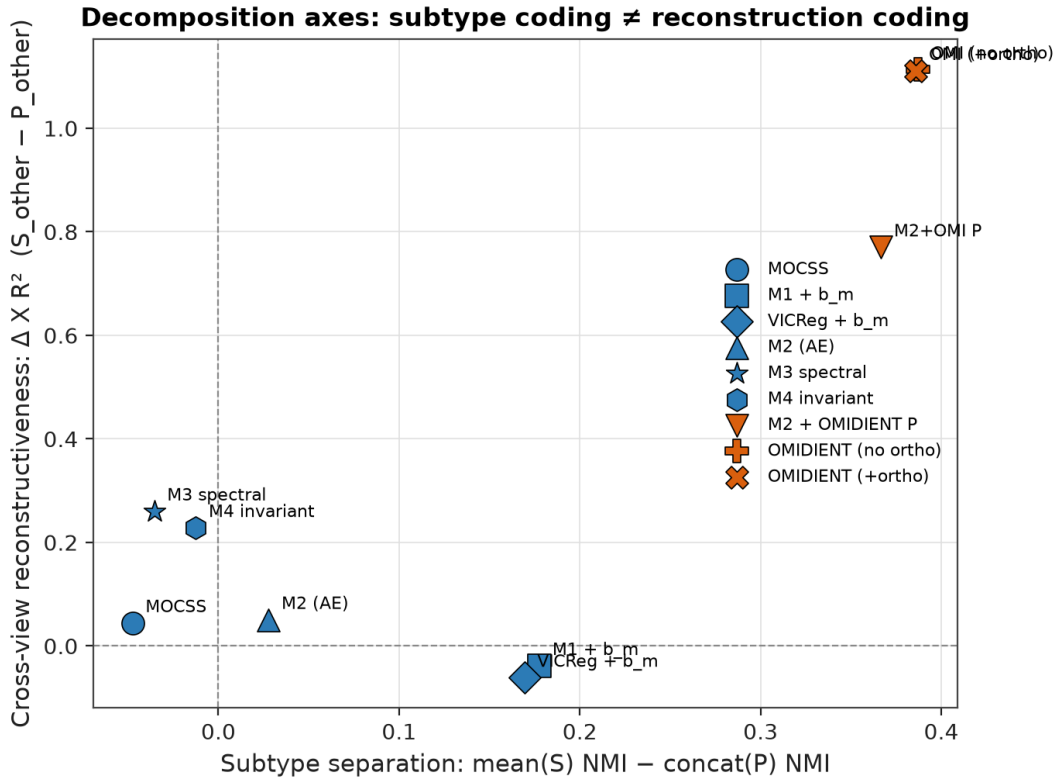


Figure 9: Subtype separation versus ΔX R^2 on BRCA. Desirable models sit toward higher separation with clearly positive ΔX R^2. Method 3 occupies the most favourable region among AE-private methods; OMIDIENT's extreme Δ coincides with private collapse and is not treated as a verified win.

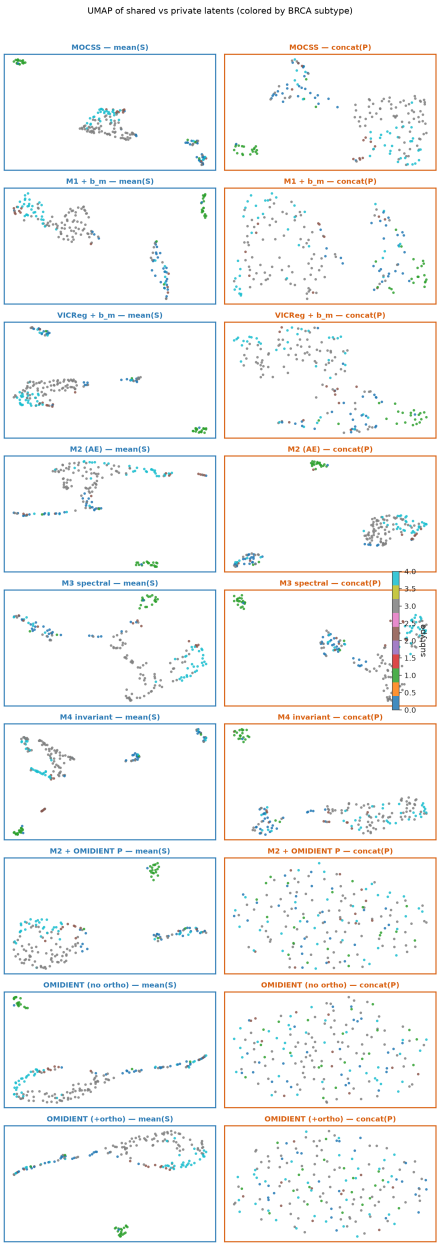


Figure 10: UMAP of mean(S) (left) and concat(P) (right) coloured by PAM50 subtype for the methods emphasised in this thesis (MOCSS, Method 1, Method 1+VICReg, Method 2, Method 3, OMIDIENT). Focus on those rows; any additional Method 4 row that may appear in the panel is outside the scope of this thesis and should be ignored. MOCSS private embeddings retain strong subtype structure; Method 2 shows structured but differently organised layouts on both sides; Method 3 preserves competitive shared geometry; OMIDIENT private embeddings lack subtype organisation consistent with collapse.

can change the NMI bars without establishing that foreign S carries the shared content that foreign P does not.

Method 1+VICReg+b_m is stabler and produces the most MOCSS-like *look* of an AE subtype split in the sense that mean(S) NMI (0.423 ± 0.021) exceeds concat(P) (0.254 ± 0.051). Yet $\Delta\text{X}\, R^2 = -0.061 \pm 0.071$ remains unfavourable and concatenation NMI (0.394 ± 0.035) still trails Method 2 and Method 3. The ablation therefore improves hygiene relative to full Method 1, but still fails the probe that defines "shared" in the cross-view sense. The lesson is explicit: low concat(P) NMI is not a sufficient statistic for disentanglement.

Method 2 versus VICReg ablation. Method 2 recovers concatenation NMI 0.434 ± 0.033 with $\Delta\text{X}\, R^2 = +0.051 \pm 0.044$. UMAP layouts are structured on both shared and private sides but organised differently, consistent with complementary rather than duplicated geometry. This profile is preferred over Method 1+VICReg: similar shared NMI, better integration, and a positive—if modest—cross-view gap. At the same time, Method 2's $\Delta\text{X}\, R^2$ remains close to MOCSS (0.045) and far below Method 3 (0.248). That gap is the clearest quantitative signal that Method 2 does *not* yet achieve the role separation this thesis aims for. Technically, the limitation is unsurprising: VICReg only soft-penalises *linear P–S* dependence (variance hinges, within-code covariance whitening, and Frobenius cross-covariance), and the joint decoder $D([P; S]) \to X$ still succeeds whenever shared and private content overlap, as long as the concatenation remains reconstructive. There is no operational pressure that forces cross-view content into S and residual content into P in the way Method 1's f/g predictors or Method 3's spectral target do. A stronger nonlinear independence criterion, or an explicit cross-view shared objective, might close part of that gap; under the present design, Method 2 behaves like a cleaner MOCSS rather than like a verified shared–private theory. Method 3 is the method that actually delivers the desired distinction on BRCA.

Method 3: clearest $\Delta\text{X}\, R^2$ win. Method 3 attains $\Delta\text{X}\, R^2 = 0.248 \pm 0.023$, a clear separation from all other AE-private methods, while remaining competitive on concatenation NMI (0.456 ± 0.025). Private concatenation NMI remains high (0.443 ± 0.014); under the interpretation rule of Chapter 4 this is acceptable when probes and UMAP indicate complementary biology rather than role collapse. The offline spectral target has unsupervised ceiling NMI(U) ≈ 0.387, so Method 3's shared NMI (0.414 ± 0.029) slightly exceeding that ceiling indicates that residual decoding, view heads, and encoder nonlinearities still contribute beyond pure Laplacian regression. Shared CKA for Method 3 is comparatively low (0.547 ± 0.017 vs. ≈ 0.76–0.79 for MOCSS/Method 2/VICReg). This is expected rather than paradoxical: the loss-best Method 3 configuration uses $\lambda_{\text{ctr}} = 0$, so the InfoNCE alignment that usually inflates pairwise shared CKA is demoted, and the spectral objective aligns $\bar{S}$ to U without requiring the individual S_m to be as linearly similar to each other as in contrastive-heavy runs. Lower CKA should therefore be read together with the strong $\Delta\text{X}\, R^2$: Method 3 trades contrastive-driven shared homogeneity for a geometry-defined shared role.

5.3 Subtyping performance

Table 5 reports standard unsupervised subtyping metrics on the loss-best configurations. MOCSS remains strong on concatenation NMI/ARI/accuracy, as expected from a model whose private codes carry subtype signal. Method 3 matches MOCSS closely on concatenation NMI (0.456 vs. 0.461) while delivering the superior $\Delta X\, R^2$ already discussed. Method 2 is slightly behind on concatenation NMI but preferable to Method 1 variants that sacrifice integration for lower private NMI. OMIDIENT concatenation NMI (0.430 ± 0.054) is competitive in mean but more variable across seeds.

Table 5: Unsupervised subtyping on BRCA for loss-best configurations (mean ± std over five seeds). Concatenation metrics use concat(P, S); s_{NMI} is NMI on mean(S).

Method	concat NMI	s_{NMI}	ARI	Acc
MOCSS	0.461 ± 0.008	0.418 ± 0.014	0.360 ± 0.032	0.643 ± 0.060
M1+b_m	0.323 ± 0.063	0.414 ± 0.030	0.190 ± 0.062	0.485 ± 0.060
M1+VICReg+b_m	0.394 ± 0.035	0.423 ± 0.021	0.273 ± 0.070	0.579 ± 0.073
Method 2	0.434 ± 0.033	0.417 ± 0.028	0.320 ± 0.049	0.621 ± 0.070
Method 3	0.456 ± 0.025	0.414 ± 0.029	0.330 ± 0.047	0.626 ± 0.049
OMIDIENT	0.430 ± 0.054	0.426 ± 0.046	—	—

Own-view and alignment complements appear in Table 6. MOCSS shows the strongest private reconstructiveness. Method 3 balances own-view $X \leftarrow S$ and $X \leftarrow P$ more evenly while keeping a large cross-view $\Delta X\, R^2$. OMIDIENT's negative $X \leftarrow P$ remains the decisive collapse indicator.

Table 6: Own-view Ridge R^2 and shared CKA on BRCA (mean ± std over five seeds).

Method	$X \leftarrow S$	$X \leftarrow P$	$X \leftarrow [P; S]$	CKA
MOCSS	0.404 ± 0.013	0.481 ± 0.006	0.363 ± 0.015	0.761 ± 0.024
M1+b_m	0.294 ± 0.032	0.414 ± 0.028	0.338 ± 0.030	0.589 ± 0.103
M1+VICReg+b_m	0.347 ± 0.040	0.427 ± 0.020	0.346 ± 0.028	0.788 ± 0.013
Method 2	0.376 ± 0.013	0.442 ± 0.004	0.377 ± 0.003	0.770 ± 0.023
Method 3	0.402 ± 0.006	0.412 ± 0.011	0.375 ± 0.010	0.547 ± 0.017
OMIDIENT	0.427 ± 0.008	-0.342 ± 0.000	0.243 ± 0.011	0.757 ± 0.037

5.4 Method-wise findings

MOCSS. Serves as a strong clustering baseline and as a negative example for role assignment: concat$(P) \approx$ concat$(P, S) >$ mean(S) with small $\Delta X\, R^2$.

OMIDIENT. Advances private-branch modelling and remains a strong integration baseline, but near-zero concat(P) NMI under the probes of this thesis coincides with private-path collapse ($X \leftarrow P < 0$); multi-seed evaluation and own-view probes are essential before reading that pattern as a verified shared–private win.

Method 1. Enforces predictability constraints effectively enough to shrink private NMI, but integration suffers and $\Delta X\, R^2$ remains unstable across seeds.

Method 1+VICReg. Improves shared–private NMI ordering relative to MOCSS and stabilises CKA, yet negative $\Delta X\, R^2$ and middling concatenation NMI make it inferior to Method 2 under the joint decision rule.

Method 2. Best simple AE-private compromise relative to Method 1 variants: joint reconstruction removes dual solo paths, VICReg replaces weak orthogonality, and multi-seed metrics remain competitive with structured UMAP on both branches. Its modest $\Delta X\, R^2$ shows that the desired cross-view role separation is still incomplete under linear VICReg alone.

Method 3. Best probe-centred outcome: largest AE-private $\Delta X\, R^2$, competitive concatenation NMI, and acceptable high private NMI interpreted as complementary signal. Lower shared CKA is explained by demoting contrastive alignment on the loss-best run. Performance remains bounded by spectral target quality ($\mathrm{NMI}(U) \approx 0.387$).

5.5 Hyperparameter sweeps

Figures 11–16 summarise label-free leaderboards over method-specific grids. Method 1 and Method 1+VICReg exhibit broad sensitivity to adversary and private-sufficiency weights: many configs that minimise private NMI also degrade concatenation NMI. Method 2 sweeps over $(\lambda_{\mathrm{rec}}, \lambda_{\mathrm{ctr}}, \lambda_{\mathrm{vic}})$ show a stable ridge of competitive models around moderate contrastive weight; the loss-best point used above is representative rather than an isolated spike. Method 3 sweeps indicate that larger spectral weights favour $\Delta X\, R^2$, while $\lambda_{\mathrm{ctr}} = 0$ frequently appears among loss-best settings, suggesting that the fused Laplacian target already supplies much of the cross-view alignment pressure. OMIDIENT orthogonality sweeps and Method 2+OMIDIENT-private sweeps confirm that near-zero private NMI and negative $X \leftarrow P$ persist across wide regulariser ranges, so collapse is not an artefact of a single λ.

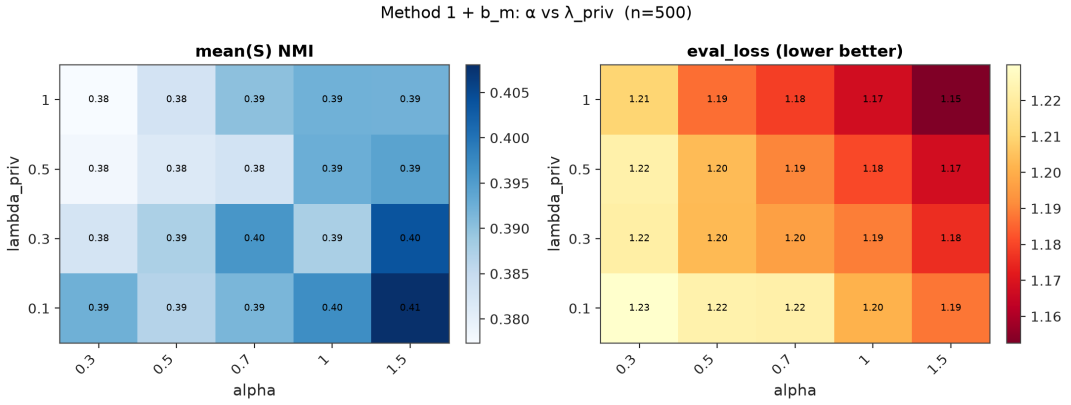


Figure 11: Hyperparameter sweep overview for Method 1 on BRCA. Performance varies sharply with predictability weights; loss-best selection does not coincide with minimal concat(P) NMI.

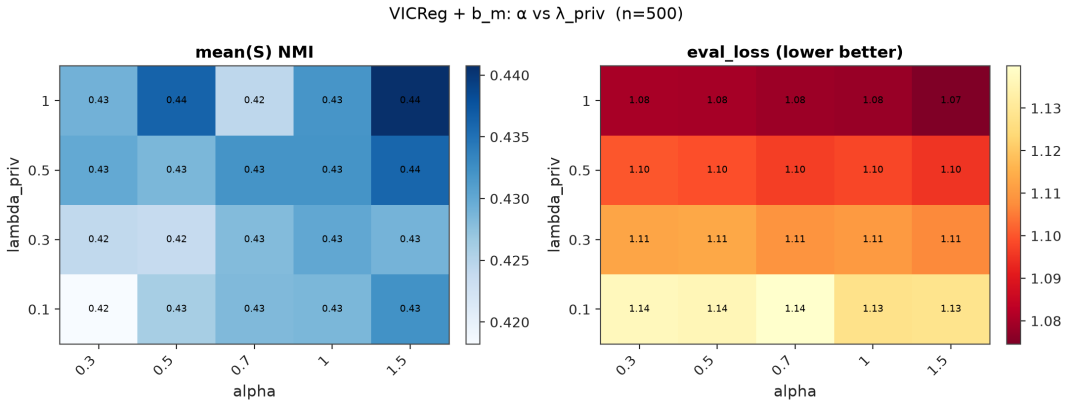


Figure 12: Hyperparameter sweep overview for Method 1+VICReg on BRCA.

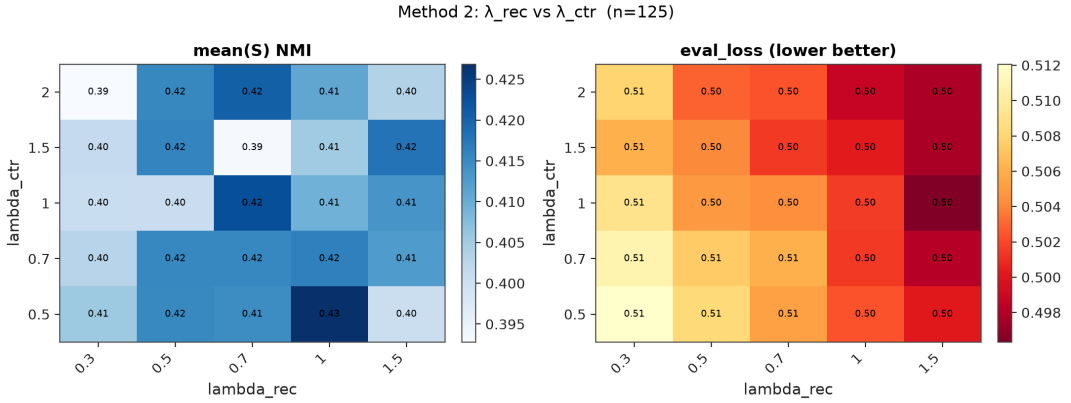


Figure 13: Hyperparameter sweep overview for Method 2 on BRCA over reconstruction, contrastive, and VICReg weights.

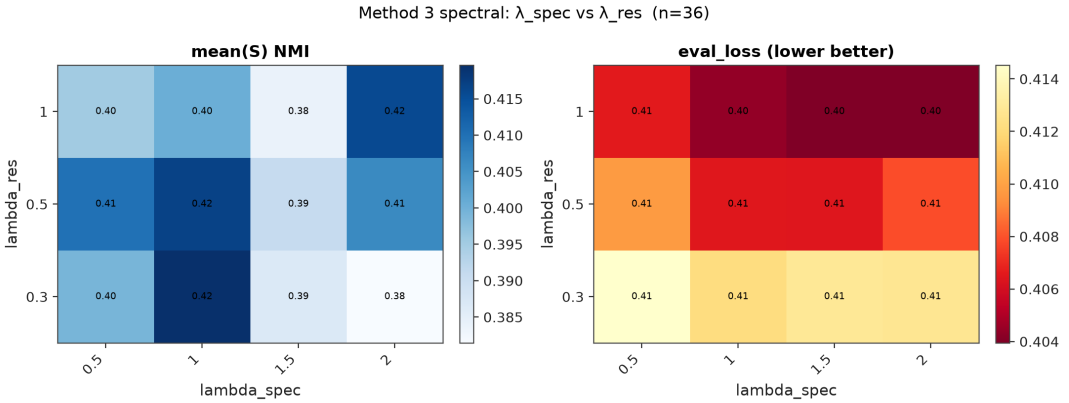


Figure 14: Hyperparameter sweep overview for Method 3 on BRCA. Stronger spectral weighting improves cross-view gaps; contrastive weight is often selected as zero under core loss ranking.

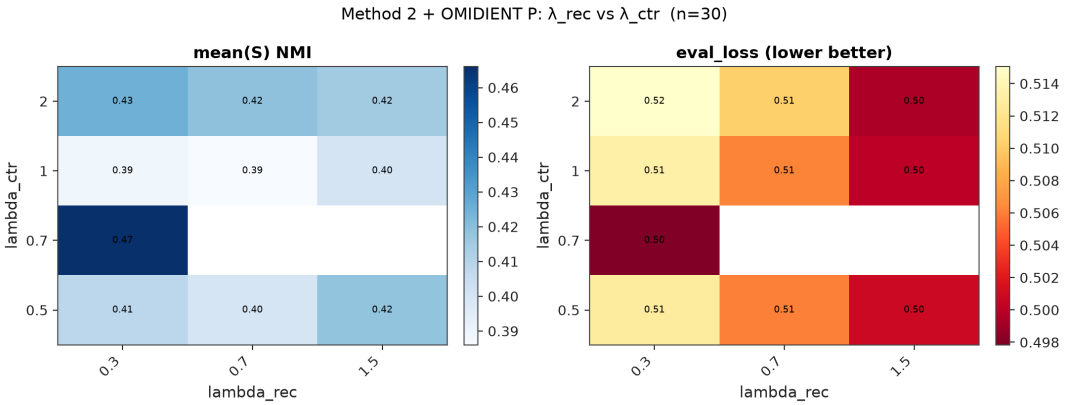


Figure 15: Sweep overview for Method 2 with OMIDIENT private codes. Private collapse signatures remain visible across the grid.

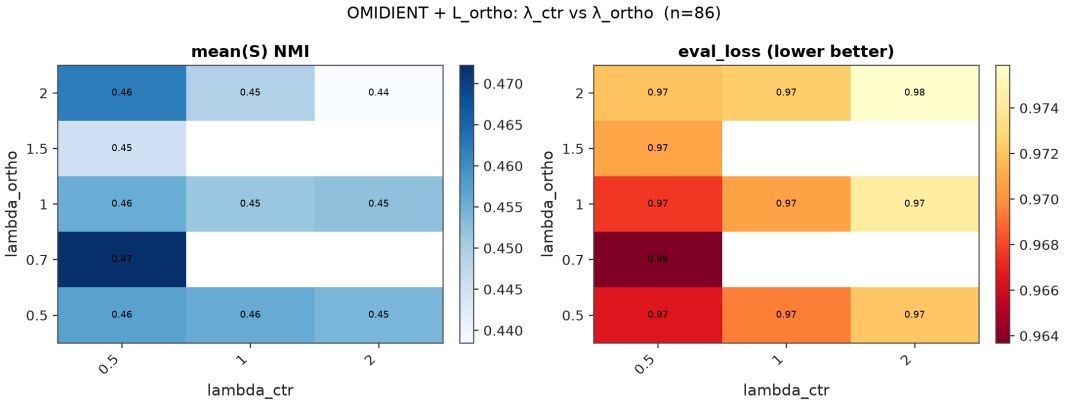


Figure 16: OMIDIENT orthogonality-weight sweep. Near-zero concat(P) NMI and negative own-view $X \leftarrow P$ are stable across λ_{ortho}.

6 Discussion

The experiments support a simple but often neglected claim: unsupervised subtype NMI of shared–private multi-omics models is not a certificate that S and P do what their names suggest. This chapter interprets the BRCA multi-seed evidence, relates it to prior work, and discusses limitations and implications.

6.1 High NMI without role fidelity

MOCSS (Chen et al., 2023) remains a strong clustering system on BRCA. Its failure mode is therefore scientifically informative rather than trivial. Because both shared and private decoders reconstruct the full input under only weak elementwise orthogonality, private codes are free to store subtype-discriminative biology. The empirical signature—concat$(P) \approx$ concat$(P, S) >$ mean(S) with $\Delta\text{X}\, R^2 \approx 0.04$ and subtype-structured private UMAP—matches that architectural incentive. Related representation-learning critiques have long warned that reconstruction objectives identify compressible features, not necessarily the factors specified in a modelling narrative (Bengio et al., 2013; Märtens and Yau, 2024). Multi-omics shared–private models are not exempt.

The practical consequence for readers of MOCSS-style papers is interpretive caution. A table that reports concatenation NMI near 0.46 answers a clustering question; it does not answer whether the shared branch is the carrier of cross-omics subtype biology. In the present multi-seed audit, private codes are at least as subtype-informative as shared codes, and own-view $X \leftarrow P$ exceeds $X \leftarrow S$. That pattern is exactly what dual solo reconstruction encourages.

OMIDIENT (Safinianaini et al., 2025) illustrates a complementary evaluation lesson rather than a dismissal of the method. OMIDIENT's contribution is a stronger private-branch model; its primary purpose was not to certify a new shared definition. Near-zero private subtype NMI and huge $\Delta\text{X}\, R^2$ look like perfect separation until own-view $X \leftarrow P$ is examined. Negative reconstructiveness indicates collapse of the Dirichlet private path under the probes of this thesis, not a verified theory of shared content. Hybridising Method 2's joint decoder with OMIDIENT private codes does not remove the collapse signature, which suggests that the Dirichlet private geometry and its KL pressure—not only MOCSS dual reconstruction—can drive uninformative P. The result also cautions against over-trusting single-seed leaderboards: favourable shared NMI in one run can shrink under seeds $\{21, \ldots, 25\}$. Technical reproducibility studies in multi-omics make the same methodological point in broader terms (Baião et al., 2025; Rappoport and Shamir, 2021).

6.2 Why minimising private NMI is the wrong objective

A natural but misleading response to MOCSS-style private leakage is to drive concat(P) subtype NMI toward zero. Method 1 and Method 1+VICReg show why that proxy fails. Both reduce private NMI relative to MOCSS, and the VICReg ablation even restores a pleasant $S > P$ NMI ordering, yet $\Delta\text{X}\, R^2$ stays near zero or negative and

concatenation NMI drops. Private codes can be made non-discriminative by collapse, by over-regularisation, or by moving information into brittle shared features that do not integrate well.

Conversely, Method 3's high private NMI is compatible with the best AE-private $\Delta X\ R^2$ when UMAP layouts differ across branches. Assay-specific correlates of subtype are biologically real (Levine et al., 2013; Parker et al., 2009); a private code that captures them without duplicating shared cross-view geometry is not a bug. Classical integrative factorisations already allowed individual structure to be scientifically meaningful (Lock et al., 2013; Argelaguet et al., 2018). Deep models should inherit that nuance rather than treating any private subtype signal as contamination.

The decision rule used in this thesis—jointly read $\Delta X\ R^2$, UMAP role separation, and integration NMI—is therefore not a cosmetic preference. It is required to distinguish complementary private biology from stolen shared signal and from collapsed privacy. Under that rule, Method 1+VICReg's attractive NMI split is insufficient, Method 2 is preferable among simple AE-private fixes, and Method 3 is the clearest overall win on role separation without surrendering competitive concatenation NMI.

6.3 What the proposed methods achieve

Method 1 operationalises shared content as cross-view predictability and private content as residual sufficiency. The idea is aligned with classical shared–individual decompositions such as JIVE and MOFA (Lock et al., 2013; Argelaguet et al., 2018), translated into deep predictors f, g, k, and b. In the BRCA AE setting studied here, the adversarial estimator proved brittle: seed-to-seed variance in $\Delta X\ R^2$ is large, and loss-best configurations under-integrate. The negative result is still useful. It shows that writing an information-theoretic story is easier than estimating it stably with fifteen networks and two-step minimax training.

Method 2 offers a more pragmatic fix: remove dual solo reconstruction, impose VICReg independence (Bardes et al., 2022), and keep contrastive alignment (Chen et al., 2020). Empirically it is the preferred AE-private alternative to the VICReg ablation of Method 1 when both probes and integration matter. The UMAP evidence that shared and private geometries are both structured but differently organised supports the complementary-biology reading of moderate private NMI. Method 2 also has a practical advantage for future extensions: its single-optimiser, all-positive objective is easier to hybridise than Method 1's adversaries, as the OMIDIENT-private companion experiment demonstrates even though that companion inherits private collapse. Nevertheless, Method 2's $\Delta X\ R^2$ (≈ 0.05) remains close to MOCSS and far below Method 3 (≈ 0.25). That is a genuine limitation of the model, not a measurement artefact: VICReg only soft-constrains linear P–S dependence, and joint reconstruction still permits overlapping content whenever the concatenated decoder succeeds. Without an explicit cross-view definition of sharedness, Method 2 improves hygiene relative to dual solo reconstruction but does not fully enforce the shared–private distinction sought in this thesis.

Method 3 goes further by giving shared codes an explicit spectral geometry target derived from fused patient neighbourhoods (Wang et al., 2014; Ng et al., 2002;

von Luxburg, 2007). That choice reconnects deep shared–private learning to the classical insight that cross-view agreement lives in sample geometry, not only in per-view feature reconstruction. The large $\Delta X\, R^2$ gain (≈ 0.25) with competitive concatenation NMI (≈ 0.46) is the strongest positive result of the thesis. Residual decoding with stop-gradient on S prevents the shared code from being rewritten as a second full autoencoder, while GRL leakage control and view heads provide additional role regularisation (Appendix A). The comparatively low shared CKA of Method 3 (≈ 0.55) is consistent with demoting contrastive alignment ($\lambda_{\text{ctr}} = 0$ on the loss-best run): spectral regression aligns $\bar{S}$ to U without forcing the same pairwise linear homogeneity that InfoNCE induces among S_m. The spectral ceiling $\text{NMI}(U) \approx 0.387$ clarifies a limitation and a research opportunity: Method 3 cannot conjure subtype signal absent from the fused graph, so improving graph construction, affinity kernels, or modality weighting is a direct avenue for further gains.

6.4 Relation to evaluation standards

Recent multi-omics benchmarking emphasises robustness, multiple metrics, and careful preprocessing disclosure (Cantini et al., 2021; Rappoport and Shamir, 2021; Baião et al., 2025; Mohr et al., 2024). This thesis adds a missing axis for the shared–private subclass of models: role probes. Reporting only mean(S) NMI can hide private leakage; reporting only concat(P) NMI can reward collapse. Cross-view $\Delta X\, R^2$ and own-view $X \leftarrow P$ together catch both failure modes in the BRCA results. Embedding visualisations remain indispensable qualitative checks, especially when numerical gaps are modest, as for Method 2 relative to MOCSS.

Label-free model selection deserves a brief remark. Ranking by core validation loss avoids leaking PAM50 into hyperparameter choice, which is appropriate for an unsupervised claim. It does not optimise $\Delta X\, R^2$ directly. Some sweep cells may therefore dominate the probe metrics while losing on loss rank. The thesis treats that mismatch as an acceptable cost of honest unsupervised selection; future work could explore probe-inspired but still label-free regularisers.

6.5 Limitations

Several limitations bound the present conclusions. First, headline results are BRCA-only; KIRC remains to be reported in a later revision, so claims about disease transfer are provisional. Second, PAM50 labels, while standard, are transcriptomic in origin and imperfect as multi-omics ground truth (Parker et al., 2009; Cantini et al., 2021). Third, VICReg enforces linear independence, not full statistical disentanglement (Märtens and Yau, 2024; Zbontar et al., 2021). Fourth, Method 1's two-step adversaries may admit stronger estimators, occupancy constraints, or single-step gradient-reversal formulations not explored here. Fifth, figure panels exported from the full repository visualisation may include rows for methods outside this thesis's scope; the narrative intentionally ignores those rows. Sixth, encoder depth and width were fixed to the MOCSS-aligned MLP backbone; capacity ablations might change absolute NMI even if relative role failures persist. Finally, biological interpretation of private

complementary signal remains correlative: the thesis does not map private dimensions to specific pathways or CpG modules.

6.6 Implications for multi-omics representation learning

For practitioners, the implication is evaluative: published shared–private models should report cross-view reconstruction gaps and own-view private reconstructiveness alongside clustering tables. A minimal reporting set suggested by this work is (i) NMI/ARI on mean(S), concat(P), and concat(P, S); (ii) $\Delta\mathrm{X}\, R^2$; (iii) own-view $X \leftarrow P$; and (iv) at least one embedding figure coloured by subtype for both branches.

For method designers, the implication is architectural. If P is meant to be residual, do not train it as a second full autoencoder of X. If S is meant to be shared, give it an explicit cross-view target—predictability, contrastive agreement, or spectral geometry—that reconstruction alone does not provide. Classical factorisations already embodied these principles in linear form (Lock et al., 2013; Shen et al., 2009; Argelaguet et al., 2018); deep models must re-enforce them rather than assume that naming latents "shared" and "private" is sufficient. Among the concrete designs studied here, Method 3 currently offers the best empirical trade-off on BRCA, with Method 2 as the simpler AE-private baseline recommended when spectral precomputation is undesirable.

7 Conclusions

This thesis examined shared–private representation learning for unsupervised multi-omics cancer subtyping, with emphasis on whether latent codes named shared (S) and private (P) actually realise that decomposition. Using TCGA BRCA multi-omics data and multi-seed evaluation over seeds $\{21, \ldots, 25\}$, the work audited MOCSS and OMIDIENT, proposed Methods 1–3 that enforce clearer role separation, and ranked models by a probe-centred protocol rather than by private-subtype NMI alone.

The first conclusion is negative but central: high unsupervised subtype NMI does not imply faithful shared–private roles. In MOCSS, private concatenation NMI approximately matches full concatenation (≈ 0.47 vs. ≈ 0.46) and exceeds mean shared NMI (≈ 0.42), while $\Delta\mathrm{X}\, R^2$ remains near 0.04 and UMAP shows subtype structure in private embeddings. In OMIDIENT, near-zero private NMI and large $\Delta\mathrm{X}\, R^2$ (≈ 1.11) coexist with collapsed own-view private reconstructiveness ($X \leftarrow P \approx -0.34$), so the apparent separation is not evidence of a verified shared theory. Single-seed optimism further shrinks under multi-seed aggregation.

The second conclusion is methodological. Models should be judged jointly by cross-view $\Delta\mathrm{X}\, R^2$, qualitative UMAP role separation, and integration NMI. Minimising concat(P) subtype NMI is not a valid primary objective: Method 1+b_m and Method 1+VICReg achieve lower private NMI (approximately 0.24–0.25) without securing stable positive $\Delta\mathrm{X}\, R^2$ or competitive concatenation performance, whereas Method 3 tolerates high private NMI (≈ 0.44) while delivering the clearest AE-private cross-view gap.

The third conclusion is constructive. Among the proposed approaches, Method 3 provides the strongest empirical support for improved shared–private distinction on BRCA ($\Delta\mathrm{X}\, R^2 \approx 0.25$ with concatenation NMI≈ 0.46), subject to the spectral ceiling of the offline fused Laplacian target (NMI(U) ≈ 0.387). Method 2 is the preferred simpler AE-private alternative to Method 1+VICReg, combining joint reconstruction, VICReg independence, and competitive multi-seed integration (concatenation NMI≈ 0.43, $\Delta\mathrm{X}\, R^2 \approx +0.05$) with structured but non-identical shared and private geometries.

KIRC experiments are reserved for a later revision. Future work should improve fused-graph targets for Method 3, harden predictability estimators beyond Method 1's two-step adversaries, test whether non-collapsing private priors can be combined with Method 2-style joint reconstruction, and extend the probe suite to additional tumour cohorts and omics combinations. More broadly, deep multi-omics models that advertise shared–private structure should treat that structure as an empirical claim to be tested, not as a consequence of architecture diagrams or clustering leaderboards.

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A Method 3 spectral details

This appendix expands the Method 3 construction summarised in Section 4. Notation matches the main text: modalities $m \in \{1, 2, 3\}$, encoders produce $(S_m, P_m) \in \mathbb{R}^d$ with $d = 32$, and $\bar{S} = (S_1 + S_2 + S_3)/3$.

A.1 Offline fused spectral target

For each modality m, let $X_m \in \mathbb{R}^{N \times d_m}$ denote the feature matrix of the N patients used to build the graph (unlabelled; no PAM50 information). Apply PCA fitted on the training split to $d_{\text{pca}} = 50$ dimensions, then ℓ_2-normalise each patient row. Build a mutual k-nearest-neighbour affinity with $k = 15$ and cosine similarity:

$$A_m[i, j] = \begin{cases} \cos(z_{m,i}, z_{m,j}), & j \in \text{kNN}_k(i) \text{ or } i \in \text{kNN}_k(j), \\ 0, & \text{otherwise}, \end{cases} \tag{A1}$$

symmetrise by $A_m \leftarrow \max(A_m, A_m^\top)$, and zero the diagonal. Fuse the three affinities by averaging,

$$A_{\text{fused}} = \frac{1}{3}\big(A_1 + A_2 + A_3\big). \tag{A2}$$

Let $D = \text{diag}\big(\sum_j A_{\text{fused}}[i, j]\big)$ and form the normalised Laplacian

$$L = I - D^{-1/2} A_{\text{fused}} D^{-1/2}. \tag{A3}$$

Compute the eigendecomposition $L\,U_{\text{full}} = U_{\text{full}}\Lambda$ and retain the $r = 16$ eigenvectors corresponding to the smallest eigenvalues *after* the trivial (near-constant) eigenvector. Row-normalise the resulting matrix $U \in \mathbb{R}^{N \times r}$. The matrix U is the offline spectral target used throughout Method 3 training. The unsupervised NMI of k-means on U itself is reported as the spectral ceiling NMI(U).

A.2 Spectral regression

A two-layer head $f_{\text{spec}} : \mathbb{R}^d \to \mathbb{R}^r$ (hidden tanh, linear output) maps the mean shared code to the spectral target:

$$\hat{U}_i = f_{\text{spec}}(\bar{S}_i), \qquad \mathcal{L}_{\text{spec}} = \frac{1}{N} \sum_{i=1}^{N} \big\|\hat{U}_i - U_i\big\|_2^2. \tag{A4}$$

This term is the operational definition of sharedness in Method 3: S is trained to match multi-omics neighbourhood geometry rather than to reconstruct assay intensities.

A.3 Residual private reconstruction

For each modality,

$$\hat{X}_m = \text{Dec}_m\big(\text{concat}(P_m, \text{stopgrad}(S_m))\big), \qquad \mathcal{L}_{\text{res}} = \sum_{m=1}^{3} \text{recon}(\hat{X}_m, X_m), \tag{A5}$$

where recon is the MOCSS-style centred, row-normalised matrix-norm metric. Stop-gradient on S_m ensures that residual decoding cannot redefine the shared code as a second full autoencoder of X_m.

A.4 View-identity head

A classifier $h_{\text{view}} : \mathbb{R}^d \to \mathbb{R}^3$ is applied to each private code:

$$\mathcal{L}_{\text{view}} = \sum_{m=1}^{3} \text{CE}\big(h_{\text{view}}(P_m),\ m-1\big). \tag{A6}$$

Minimising $\mathcal{L}_{\text{view}}$ encourages P_m to retain assay-identifying information, i.e. content that is useful for recognising the modality and therefore less likely to be purely shared biology.

A.5 GRL non-leakage adversary

Let q_i be a soft cluster assignment derived from stop-gradient features of $\bar{S}$ (refreshed cluster prototypes / soft labels with K clusters). An adversary h_{adv} is applied to private codes through a gradient-reversal layer GRL_λ:

$$\mathcal{L}_{\text{leak}} = \sum_{m=1}^{3} \text{CE}\big(h_{\text{adv}}(\text{GRL}_\lambda(P_m)),\ q\big). \tag{A7}$$

Forward evaluation trains h_{adv} to predict q from P_m; reversed gradients encourage the encoder to make P_m uninformative about the shared soft clustering. This is the non-leakage regulariser that complements residual reconstruction.

A.6 Full Method 3 objective

Optional VICReg and demoted contrastive terms may be included. The training objective used in the BRCA sweeps is

$$\mathcal{L} = \lambda_{\text{spec}}\,\mathcal{L}_{\text{spec}} + \lambda_{\text{res}}\,\mathcal{L}_{\text{res}} + \lambda_{\text{ctr}}\,\mathcal{L}_{\text{ctr}} + \lambda_{\text{vic}}\,\mathcal{L}_{\text{vic}} + \lambda_{\text{view}}\,\mathcal{L}_{\text{view}} + \lambda_{\text{leak}}\,\mathcal{L}_{\text{leak}}. \tag{A8}$$

Label-free ranking uses the core score $\mathcal{L}_{\text{spec}} + \mathcal{L}_{\text{res}} + \mathcal{L}_{\text{ctr}}$ (unweighted), treating VICReg, view, and leak terms as training regularisers. The loss-best BRCA configuration reported in the main text uses $\lambda_{\text{ctr}} = 0$, which helps explain the comparatively low shared CKA of Method 3.